Norm-fiber densities and relative traces in characteristic two

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Abstract

Let $t = 2^s$, $s \geq 2$, and let $C_d \leq \mathbb{F}_{t^4}^\times$ have order $d = (t+1)(t^2+1)$. We estimate the populations of linked elements $a, 1+a \in C_d$ that jointly generate C_d or each have order d. Genus-one norm fibers, an $8\sqrt{t}$ mixed-character bound and exact averaging of the norm-dependent coefficients give

$$|\mathcal{R}_d| = (t+1)^2 \prod_{p|d}(1-p^{-2}) + O\big(t^{3/2}2^{\omega(d)}\big).$$

The simultaneous full-order population has main term $(\varphi(d)/d)^2(t+1)^2$ and an explicit error. For every $c \in \mathbb{F}_{t^2}^*$, a four-point character bound estimates the order-d elements with $a + a^{t^2} = c$ by $(\varphi(d)/d)(t+1) + O((\varphi(d)/d)2^{\omega(d)}\sqrt{t})$, uniformly in c. A prime sieve, an inverse-support reduction and exact representative certificates classify all empty nonzero slices: they form the subgroup of order 9 at $s = 3$ and the subgroup of order 11 at $s = 5$. A cyclic convolution computes the mean within each fixed-norm class of traces, gives a variance bound involving $2^{2\omega(t^2+1)}$, and explains the exceptional populations. Scaling transfers the classification to arbitrary nonzero norm fibers, including ambient primitive elements when the prescribed norm is primitive. These refinements of classical prescribed-order statistics also yield $\mathrm{ord}(a) = \mathrm{ord}(1+a) = d$ for all $s \geq 2$. We determine the group generated by an affine complete mapping f and its prescribed companion $(\mathrm{id}+f)^{-1}$, and count its scalar realizations. General index bounds and a norm obstruction locate the optimal uniform index exponent at $1/4$, with a logarithmic gap in the sufficient range.

1 Introduction

Let C_d be the multiplicative subgroup of order $d > 1$ in a field $\mathbb{F}_q$ of characteristic two. How often do linked elements $a, 1+a \in C_d$ generate C_d, and how often do they both have order d? These conditions define different populations:

$$\mathcal{R}_d = \{a \in C_d : 1+a \in C_d,\ \langle a, 1+a\rangle = C_d\}, \qquad D_d = \#\{a : \mathrm{ord}(a) = \mathrm{ord}(1+a) = d\}.$$

The distinction matters in affine group realization. For a complete mapping f, its prescribed companion is $F(f) = (\mathrm{id}+f)^{-1}$. If $q = 2^n$ and $\mathrm{ord}_d(2) = n$, precisely $(q-1)|\mathcal{R}_d|$ scalar affine maps generate $H_d = \mathbb{F}_q^+ \rtimes C_d$ with their companion (Corollary 2.5). Counting the pair therefore gives the exact number of scalar realizations of this primitive group.

The quartic norm family

The central case has index $t-1$ in $\mathbb{F}_{t^4}^\times$, where $t = 2^s$ and $s \geq 2$. Put

$$K = \mathbb{F}_t \subset E = \mathbb{F}_{t^2} \subset L = \mathbb{F}_{t^4}, \qquad A = t+1, \quad B = t^2+1, \quad d = AB.$$

Here $C_d = \ker N_{L/K}$. Write $\theta(d) = \varphi(d)/d$, $\theta_2(d) = \prod_{p|d}(1-p^{-2})$ and $W(d) = 2^{\omega(d)}$. We prove

$$|\mathcal{R}_d| = \theta_2(d)(t+1)^2 + O(t^{3/2}W(d)),$$
$$D_d \sim \theta(d)^2(t+1)^2, \qquad D_d > 0 \quad (s \geq 2).$$

Theorems 4.3 and 4.4 give explicit errors. For joint generation the local obstruction at a prime $p \mid d$ is that both elements lie in C_d^p; for double full order, each element must avoid C_d^p. This accounts for the different factors $1-p^{-2}$ and $(1-p^{-1})^2$.

Adjacent prescribed-order pairs are classical. In particular, D_d is the $(t-1)$-primitive-pair count in [13, Proposition 3.1], and is a special case of the norm-constrained count in [14, Definition 3.1]. The center of the estimate in [13], after substitution, is $\theta(d)^2t^4/(t-1)^2$; its difference from our center is only $\theta(d)^2(2t^2-1)/(t-1)^2$. The substantive issue is therefore the error and its uniformity as t grows. General ambient-field character bounds have square-root scale t^2, already the scale of the whole norm intersection. The norm fibers recover cancellation at the smaller base-field scale.

From a fiber estimate to two densities

We first fix $b = N_{L/E}(a)$ in $U = \ker N_{E/K}$, of order A. For $b^3 \neq 1$ the resulting fiber is a genus-one curve. Its population has main term $t+1$, and its eight boundary points have no K-rational points. Characters factoring through the fixed norm can be constant even when nontrivial. Proposition 3.4 isolates these exceptions and bounds the remaining mixed characters by $8\sqrt{t}$. Its proof identifies their finite-field traces, verifies geometric monodromy by boundary valuations, and controls the tame conductor.

The joint-generation coefficient on this fiber is $\lambda_A(b)\theta_2(B)$, which varies with b. Lemma 4.2 computes its average exactly as $\theta_2(d)$. Summing over all A fibers, with $b^3 = 1$ treated separately, produces the quadratic main term. Applying individual primitive indicators to the same geometric estimate gives the double full-order density.

Relative traces: a uniform distribution and two exceptional groups

For $c \in E^*$ let M_c count order-d elements with $a + a^{t^2} = c$. This is a specialization of the classical prescribed-order and prescribed-trace counts [22, 23, 10]; the quantitative issue is the error at the smaller base-field scale. Write $c = hu$ with $h \in K^*$ and $u \in U$. The reciprocal-trace change of variable $a = c/(1+\gamma)$ reduces, up to a character factor of absolute value one, to the smooth genus-one curve

$$(x+1)^2(y+1)^2 = h^4xy$$

with four geometric boundary points and inherited torus descent. Theorem 5.1 proves, uniformly in c,

$$|M_c - \theta(d)(t+1)| \leq \theta(d)(4W(d)-2)\sqrt{t}.$$

It also separates the primitive-indicator error from the exact unfiltered population, whose fluctuation is a classical binary Kloosterman sum (Lemma 5.2). A prime sieve then gives $M_c > 0$ for every $c \neq 0$ when $s \geq 10$, and also when $s = 8$. Exact certificates in the remaining small fields complete the classification: the empty nonzero traces are precisely $c^9 = 1$ at $s = 3$ and $c^{11} = 1$ at $s = 5$. These are subgroups of sizes 9 and 11; each meets K^* only at 1. For the finite range, an elementary inverse-support reduction uses the fact that $t+1$ has at most two distinct prime divisors there. It replaces all-trace coverage by 755 representative witnesses and 731 support-separation witnesses. The exceptional groups arise from an empty support at $t = 8$ and from support confined to one

prime-coordinate kernel at $t = 32$. The two-prime restriction and the separation condition are essential; they are not asserted for arbitrary supports.

Within each fixed-norm class hU, the populations form a cyclic convolution with the primitive indicator of U. Proposition 5.5 computes their mean exactly and bounds their variance by

$$16\theta(A)(1-\theta(A))\theta(B)^2W(B)^2t.$$

Unlike the pointwise estimate, this bound has no $W(A)$ factor. The same convolution explains the complete $0, 20, 18$ profile on $U = C_{33}$ at $t = 32$ (Corollary 5.6). Scaling by K^* transfers the classification to every prescribed nonzero K-norm. In particular it classifies primitive elements of L with that norm and a prescribed E-valued relative trace when the norm is primitive (Corollary 5.7). The Fourier and scaling steps are direct consequences of the geometric bound and classification, with no separate claim of a new method.

For $c = 1$, the classical substitution [17, §4] couples the two orders through $1 + a = a^{t^2}$. Its parameter count $\Delta_t = M_1$ is exactly $N_{t^4}(B, A)$ in [17, Theorem 1.2]. Theorem 5.4 shows that Δ_t is positive precisely for $s \notin \{3, 5\}$. The two zeros have different polynomial order obstructions; they are not zeros of the unfiltered slice. Consequently only two exceptional field certificates are needed to conclude $D_d > 0$ for every $s \geq 2$. The other nonzero traces describe relative-trace distributions, not further adjacent-pair counts.

Position among earlier results

Freeness indicators and prescribed-order sieves are established tools [7, 8, 9, 13]. The particular general sufficient conditions in [7, 13, 14] fail after substitution of the present parameters. Supplementary Section S7 gives the substitutions, checks the stronger prime-sieve criteria, and distinguishes that failure from any assertion about all possible refinements. The effective fiber estimate, its norm-dependent coefficients, their exact average, the uniform relative-trace theorem and its exceptional-parameter classification constitute the quantitative results established here.

The relative-trace statistic is also a line count in [6]. The product criterion of [25] fails for every permitted decomposition of the growing index $t - 1$; Appendix D gives a uniform proof of this fact and credits their classification for the $s = 2$ case. The hypersurface estimates in [26] concern related primitive-coordinate counts. Their displayed results do not impose the conjugacy relation required in the relative-trace slices studied here.

The prescribed-trace and inverse-trace counts in [27] also contain M_c through an explicit rational-image specialization and summation. Their general and prime-sieve criteria cannot hold in extension degrees two or four. Appendix D gives the exact comparison, including the repeated-factor and inverse-trace conditions. It also checks the earlier function-field criterion of [10], retaining the exact Artin–Schreier genus and allowing every squarefree sieve core.

There are also close geometric predecessors. Hachenberger's quartic primitive-normal bound [11, Theorem 1] is an aggregate of the relative-trace populations; it can be positive when a specified trace slice is empty (Appendix D). His later model [12, pp. 94, 97, 107] identifies the norm-one representatives of primitive projective points and their trace-zero oval sections. The trace-one family here additionally fixes a scale-invariant norm-to-trace ratio. The cited incidence and primitive-normal counts do not determine this affine slice; Supplementary Section S7 gives the exact comparison. Ordinary cyclotomic intersections, including the quadratic root-count parametrization, are treated in [15]. The character bound and Lang construction are classical [5, 4]; the geometric verification below makes their application effective on these fibers. Closely related subgroup-coset trace estimates in cubic norm tori occur in [20]; their characteristic assumptions exclude this binary quartic

family. Supplementary Section S7 also discusses the related prescribed-order, prescribed-norm and prescribed-trace framework of [21].

Affine groups and the uniform index range

For every affine complete mapping $f(x) = Mx + c$, we prove

$$\langle f, F(f)\rangle = \mathbb{F}_2[M]c \rtimes \langle M, (I+M)^{-1}\rangle.$$

The formula includes reducible and nonsemisimple M and yields the action criteria in Corollary 2.4. The framework comes from complete mappings and Latin squares [1]; $F^3 = \mathrm{id}$ is the classical cyclic coordinate-permutation symmetry of a transversal [2]. The group of this prescribed pair is distinct from the groups generated by larger collections of complete mappings [18, 19].

For general index $k = (q-1)/d$, Jacobi sums and divisor inversion give a uniform realization range $k \le q^{1/4}/\sqrt{\log_2 q}$ for $n \ge 16$. A primitive norm-obstructed family occurs at $k = q^{1/4} + 1$, whereas the quartic results give realization at $k = q^{1/4} - 1$. These families locate the optimal uniform exponent at $1/4$; the logarithmic gap in the sufficient range remains. Slowly varying indices already occur in [6].

Section 2 proves the group reduction and exact sieve. Sections 3–5 develop the fiber estimate, densities and all-parameter existence. Section 6 gives the general bounds and obstruction. The appendices record finite certificates, verification scope and explicit comparisons with earlier results. Coordinate Singer examples, oriented spreads, nonlinear companions and further scalar statistics appear in the supplement, whose numbers carry an S prefix.

2 Affine companions and the scalar realization problem

2.1 Complete mappings and their companions

Let $V = \mathbb{F}_2^n$, with addition denoted by $+$ or $\oplus$. A permutation f is a *complete mapping* if $(\mathrm{id} \oplus f)(x) = x \oplus f(x)$ is also a permutation. Set $F(f) = (\mathrm{id} \oplus f)^{-1}$. Function powers denote composition; orders of field elements are multiplicative orders. Composition is read from right to left.

Lemma 2.1. *Every complete mapping f of an additive group of exponent two has exactly one fixed point, namely $F(f)(0)$.*

Proof. The equation $f(x) = x$ is equivalent to $x \oplus f(x) = 0$. Since $\mathrm{id} \oplus f$ is bijective, its unique solution is $F(f)(0)$. □

Lemma 2.2. *On complete mappings of any additive group of exponent two, F is defined at every iterate and satisfies*

$$F^2(f) = \mathrm{id} \oplus f^{-1}, \qquad F^3(f) = f.$$

Proof. Put $g = (\mathrm{id} \oplus f)^{-1}$. The identity $x = g(x) \oplus f(g(x))$ gives

$$\mathrm{id} \oplus g = f \circ g.$$

Thus g is complete. Taking inverses yields

$$F^2(f) = (f \circ g)^{-1} = g^{-1} \circ f^{-1} = (\mathrm{id} \oplus f) \circ f^{-1} = \mathrm{id} \oplus f^{-1}.$$

Consequently $\mathrm{id} \oplus F^2(f) = f^{-1}$ and $F^3(f) = f$. Only addition of exponent two and bijectivity were used. □

The transversal $\{(x, f(x), x + f(x)) : x \in V\}$ is unchanged as a family when its coordinates are permuted. Its cyclic rotation induces F; Supplementary Proposition S3.1 gives this classical interpretation explicitly. For a scalar affine map $f(x) = \alpha x \oplus c$, put $\beta = (1 \oplus \alpha)^{-1}$ and $\kappa = 1 \oplus \alpha^{-1}$. Then $F(f)(x) = \beta x \oplus \beta c$ and $F^2(f)(x) = \kappa x \oplus \alpha^{-1} c$, so the slopes follow $\alpha \mapsto \beta \mapsto \kappa \mapsto \alpha$, with $\alpha\beta\kappa = 1$. The exponent-two hypothesis matters: over $\mathbb{F}_5$, replacing $\oplus$ by ordinary field addition $+$, all three iterates of $f(x) = 2x + 1$ are defined, but $F^3(f)(x) = 2x + 2 \neq f(x)$.

2.2 The generated affine group

For a vector u write $\tau_u(x) = x \oplus u$. If $U \leq V$ is invariant under $M \leq \mathrm{GL}(V)$, we realize $U \rtimes M$ as $\{x \mapsto Lx \oplus u : L \in M,\ u \in U\}$ in the given coordinates. Composition has the usual rule $(u, L)(v, N) = (u \oplus Lv, LN)$.

Theorem 2.3 (Affine companion group)**.** *Let $V = \mathbb{F}_2^n$, $n \geq 1$, and let $f(x) = Ax \oplus c$ be complete, so $A, I + A \in \mathrm{GL}(n, 2)$. Put*

$$B = (I + A)^{-1}, \qquad R = \mathbb{F}_2[A], \qquad M_A = \langle A, B \rangle,$$
$$U = Rc = \mathrm{span}_{\mathbb{F}_2}\{c, Ac, \ldots, A^{n-1}c\}.$$

For every $c \in V$, including $c = 0$,

$$\langle f, F(f) \rangle = U \rtimes M_A, \qquad |\langle f, F(f) \rangle| = 2^{\dim U} |M_A|.$$

In particular its pure translations are exactly $\{\tau_u : u \in U\}$.

Proof. The minimal polynomial of A is coprime to t and $t + 1$. Bezout identities therefore show that A^{-1} and B belong to R; their inverses do as well. Thus $M_A \leq R^\times$ is abelian and preserves U. Cayley–Hamilton gives the displayed finite spanning set for U. Since $g = F(f)$ has the explicit form $g(x) = Bx \oplus Bc$, both generators lie in $U \rtimes M_A$, proving one inclusion.

Using $AB = I + B$ and $[f, g] = f \circ g \circ f^{-1} \circ g^{-1}$, direct composition gives

$$g \circ f(x) = ABx, \qquad f \circ g(x) = ABx \oplus Bc, \qquad [f, g] = \tau_{Bc}.$$

Conjugation by f sends τ_u to τ_{Au}. Hence the generated group contains all translations whose vectors lie in $\mathrm{span}\{A^j Bc : j \geq 0\} = RBc = Rc = U$, where B is a unit of R. It contains $\tau_c \circ f = A$ and $\tau_{Bc} \circ g = B$ as pure linear maps, so it contains M_A as well. This proves the reverse inclusion. Distinct pairs (u, L) give distinct affine maps, yielding the order and translation statements. □

Corollary 2.4 (Action criteria)**.** *In Theorem 2.3, write $H = \langle f, F(f) \rangle$. Then H is transitive on V if and only if $U = V$, equivalently c is a cyclic vector for A. It is primitive if and only if $c \neq 0$ and the characteristic polynomial of A is irreducible. If A is identified with multiplication by $\alpha \in \mathbb{F}_{2^n}$ in this irreducible case, then for $c \neq 0$*

$$|H| = 2^n \,\mathrm{lcm}\big(\mathrm{ord}(\alpha), \mathrm{ord}(1 \oplus \alpha)\big).$$

It is doubly transitive if and only if $c \neq 0$, A is irreducible, and this least common multiple is $2^n - 1$. In that case $H = \mathrm{AGL}(1, 2^n)$ in these field coordinates.

Proof. The orbit of 0 is exactly U, giving transitivity. When $U = V$, the group contains the full regular translation group and its stabilizer of 0 is M_A. A block containing 0 is an additive subspace: if u belongs to the block, its translate by u meets the block at u, so the block property

makes that translate equal to the block. The block is also M_A-invariant. Conversely the cosets of any M_A-invariant subspace form a block system (cf. [16]). An A-invariant subspace is invariant under the polynomial maps B, A^{-1}, B^{-1}, so the M_A-invariant subspaces are exactly the A-invariant ones. The action is therefore primitive precisely when A acts irreducibly; this is equivalent to an irreducible characteristic polynomial. Such an A makes every nonzero vector cyclic, which explains the condition $c \neq 0$.

An irreducible A identifies V with $\mathbb{F}_{2^n}$ and A with multiplication by α; the two multipliers generate $\langle \alpha, (1 \oplus \alpha)^{-1} \rangle$ in the cyclic multiplicative group. Its order is the least common multiple of the two element orders. Double transitivity means that M_A is transitive on $V \setminus \{0\}$, which holds exactly when this subgroup is the entire multiplicative group. Double transitivity implies primitivity, so the conditions are also necessary outside the irreducible case. □

2.3 Scalar realizations and an exact arithmetic sieve

Corollary 2.5 (Realizing prescribed scalar affine subgroups)**.** *Fix $n \geq 2$, put $q = 2^n$, and regard $\mathbb{F}_q$ as an n-dimensional vector space over $\mathbb{F}_2$. For a divisor $d > 1$ of $q - 1$, let C_d be the unique subgroup of order d of $\mathbb{F}_q^\times$, and define*

$$\mathcal{R}_d = \{a \in C_d : \ 1 + a \in C_d, \ \langle a, 1 + a \rangle = C_d\}.$$

The particular affine subgroup $H_d = \mathbb{F}_q^+ \rtimes C_d \leq \mathrm{AGL}(1, q)$ is generated by f and $F(f)$ for a scalar affine complete mapping $f(x) = ax + c$ if and only if

$$\mathrm{ord}_d(2) = n \quad \textit{and} \quad \mathcal{R}_d \neq \varnothing.$$

Here $\mathrm{ord}_d(2)$ denotes the multiplicative order of 2 modulo the odd integer d. When the conditions hold, the exact number of scalar affine maps f realizing this specified subgroup H_d (in these fixed field coordinates) is

$$(q - 1)|\mathcal{R}_d|.$$

Proof. If $\langle f, F(f) \rangle = H_d$, Theorem 2.3 shows that its linear part is $\langle a, (1+a)^{-1} \rangle = C_d$, so $a \in \mathcal{R}_d$. Its pure translations fill the field, which forces $c \neq 0$ and $\mathbb{F}_2[a] = \mathbb{F}_q$. Since $C_d \leq \mathbb{F}_q^\times$ and $\langle a, 1 + a \rangle = C_d$, the least subfield containing C_d is $\mathbb{F}_2[a]$. The subfield generated by a multiplicative subgroup of order d has degree $\mathrm{ord}_d(2)$ over $\mathbb{F}_2$; thus this order is n.

Conversely, suppose $\mathrm{ord}_d(2) = n$ and $a \in \mathcal{R}_d$. Because $1 + a \in \mathbb{F}_2[a]$ and $C_d = \langle a, 1 + a \rangle$, the field $\mathbb{F}_2[a]$ contains C_d, hence equals $\mathbb{F}_q$. In particular $a \neq 0, 1$, multiplication by a is irreducible over $\mathbb{F}_2$, and for any $c \neq 0$ one has $\mathbb{F}_2[a]c = \mathbb{F}_q$. Theorem 2.3 now gives $\langle f, F(f) \rangle = H_d$. There are exactly $|\mathcal{R}_d|$ admissible slopes and $q - 1$ admissible translations; different pairs (a, c) define different maps. □

Scope. This is an exact realizability test for specified scalar affine subgroups, not a classification of all subgroups of $\mathrm{AGL}(n, 2)$ up to conjugacy. The remaining arithmetic question is to determine the nonemptiness and size of $\mathcal{R}_d$ uniformly in n, d; the criterion by itself is a consequence of Theorem 2.3.

Corollary 2.6 (An arithmetic sieve for realizable slopes)**.** *Retain the notation of Corollary 2.5, and for every divisor e of $q - 1$ set*

$$T_e = |\{a \in C_e : 1 + a \in C_e\}|.$$

Then the number of slopes realizing C_d satisfies

$$|\mathcal{R}_d| = \sum_{e|d} \mu(d/e) T_e, \tag{2.1}$$

where μ is the number-theoretic Moebius function. In fact,

$$T_e = \deg \gcd_{\mathbb{F}_2[t]}\big(t^e + 1, (t+1)^e + 1\big), \tag{2.2}$$

so these numbers depend on e, not on the ambient field containing C_e. Moreover, for $d > 3$ one has

$$|\mathcal{R}_d| \equiv 0 \pmod 3;$$

for $d = 3$ (necessarily n even), $|\mathcal{R}_3| = 2$. In particular, the full-translation subgroup $\mathbb{F}_{2^n}^+ \rtimes C_3$ is realizable by a scalar affine companion pair in fixed field coordinates exactly when $n = 2$.

Proof. Every pair $(a, 1+a)$ counted by T_e generates a unique subgroup $C_s \le C_e$. Therefore $T_e = \sum_{s|e} |\mathcal{R}_s|$, extending the definition by $\mathcal{R}_1 = \varnothing$. Moebius inversion yields (2.1). Since $e \mid q-1$, both polynomials in (2.2) split into simple linear factors over $\mathbb{F}_q$ (their derivatives are nonzero, as e is odd). Their common roots are precisely the elements a counted by T_e, proving the polynomial-gcd formula.

The fractional-linear transformation $\varphi(a) = (1+a)^{-1}$ preserves $\mathcal{R}_d$: indeed $\varphi(a)$ and $1 + \varphi(a) = a/(1+a)$ belong to C_d and generate exactly the subgroup generated by a and $1+a$. Direct simplification gives $\varphi^3(a) = a$. A fixed point obeys $a^2 + a + 1 = 0$, hence has multiplicative order three and can lie in $\mathcal{R}_d$ only when $d = 3$. Thus for $d > 3$ all φ-orbits in $\mathcal{R}_d$ have length three. For $d = 3$, the two nonidentity elements of $\mathbb{F}_4^\times$ comprise $\mathcal{R}_3$. Finally $\mathrm{ord}_3(2) = 2$, so Corollary 2.5 gives the last claim. □

As a symbolic nonrealizability example in dimension six,

$$\gcd(t^9 + 1, (t+1)^9 + 1) = t^2 + t + 1,$$

while the analogous gcd for exponent 21 is

$$(t^2 + t + 1)(t^3 + t + 1)(t^3 + t^2 + 1).$$

Together with the divisor-sum formula, these identities give $|\mathcal{R}_9| = 2 - 2 = 0$ and $|\mathcal{R}_{21}| = 8 - 6 - 2 = 0$. Although $\mathrm{ord}_9(2) = \mathrm{ord}_{21}(2) = 6$, neither $\mathbb{F}_{64}^+ \rtimes C_9$ nor $\mathbb{F}_{64}^+ \rtimes C_{21}$ can be realized by a scalar affine companion pair. These are exact polynomial identities, not claims of a complete nonrealizability classification.

The quantities T_e are standard cyclotomic intersection counts. For example, if $k = (q-1)/e$ and χ is a multiplicative character of order k on $\mathbb{F}_q$, extending all powers (including the trivial character) by zero at zero, then

$$T_e = \frac{1}{k^2} \sum_{i,j=0}^{k-1} \sum_{x \in \mathbb{F}_q} \chi^i(x) \chi^j(1+x).$$

This identity is simply the multiplicative-character indicator of C_e. Accounting separately for the trivial-character terms gives the effective classical estimate in Proposition 6.1; its combination with a maximal-subgroup sieve will prove that $\mathcal{R}_d$ is nonempty in the families below.

3 Genus-one norm fibers and mixed characters

3.1 The quartic tower and its fibers

Throughout Sections 3–5, put

$$t = 2^s \ (s \ge 2), \quad K = \mathbb{F}_t, \quad E = \mathbb{F}_{t^2}, \quad L = \mathbb{F}_{t^4}, \quad A = t + 1, \quad B = t^2 + 1, \quad d = AB.$$

Here $\gcd(A,B)=1$ and $C_d=\ker N_{L/K}$. Write $U=\ker N_{E/K}=C_A$ and $V=\ker N_{L/E}=C_B$; in these sections V denotes this cyclic norm-one group. The ambient multiplicative index is $(t^4-1)/d=t-1$. The aim is to count joint generation and, separately, simultaneous full order for a and $1+a$. The following elementary quadratic reduction is the characteristic-two norm root-count calculation; compare [15, Proposition 3.6] with base field E. Its role is to expose the fibers to be estimated.

Proposition 3.1 (Quadratic parametrization)**.** *The elements $a\in L\setminus E$ with $a,1+a\in C_d$ correspond, two at a time, to the ordered pairs $(b,w)\in U^2$ satisfying*

$$c=1+b+w\neq 0,\qquad \mathrm{Tr}_{E/\mathbb{F}_2}(b/c^2)=1.$$

The two elements are the roots of X^2+cX+b, and $N_{L/E}(a)=b$, $N_{L/E}(1+a)=w$. The remaining elements, in E, are precisely the two roots of X^2+X+1 when s is odd, and there are none when s is even. These remaining elements generate only C_3.

Proof. If $a\notin E$, write $b=a^{t^2+1}$ and $c=a+a^{t^2}$. Then $c\neq 0$, the minimal polynomial is X^2+cX+b, and $(1+a)^{t^2+1}=1+b+c=w$. Norm transitivity puts b,w in U. After division by c^2, the quadratic is an Artin–Schreier equation; it is irreducible exactly when the displayed absolute trace is one. Conversely, that trace condition makes the quadratic irreducible over E, with both roots in $L\setminus E$. Their two relative norms are b,w, so norm transitivity gives $a,1+a\in C_d$. The relative norms determine the ordered pair uniquely.

For $a\in E$, the same norm conditions reduce to $a,1+a\in U$, since $N_{L/K}(a)=N_{E/K}(a)^2$. Thus $a^t=a^{-1}$ and $(1+a)^t=(1+a)^{-1}$. Substitution gives $(1+a)(1+a^{-1})=1$, or $a^2+a+1=0$. Such roots lie in U exactly when $3\mid t+1$, equivalently when s is odd. Their two norms satisfy $1+b+w=0$. In particular, the omitted $c=0$ branch contributes no realizing slope. □

Lemma 3.2 (A genus-one norm fiber)**.** *Fix $b\in U$ with $b^3\neq 1$ and put*

$$X_b(K)=\{a\in L: N_{L/E}(a)=b,\ N_{L/K}(1+a)=1\}.$$

This set is the full set of K-points of a smooth projective geometrically connected curve C_b/K of genus one. Over $\overline{K}$ the affine part has equation

$$(x+1)(x+b)(y+1)(y+b^{-1})=xy\qquad (x,y\neq 0). \tag{3.1}$$

Its completion in $\mathbb{P}^1\times\mathbb{P}^1$ has eight boundary points, none rational over K. Consequently

$$\bigl|\#X_b(K)-(t+1)\bigr|\leq 2\sqrt{t}.$$

Proof. The relative-norm equation defines a variety over K by restriction of scalars. In the four split coordinates of $\mathrm{Res}_{L/K}\mathbb{G}_m$, write $(a_0,a_1,a_2,a_3)=(x,y,b/x,b^{-1}/y)$. The remaining norm equation is exactly (3.1). Let $P=(x+1)(x+b)$ and $Q=(y+1)(y+b^{-1})$. At a singular point in the torus, $PQ=xy\neq 0$ and

$$(b+1)Q+y=0,\qquad (b^{-1}+1)P+x=0.$$

Multiplication would force $b+b^{-1}=1$, hence $b^3=1$. This contradicts the hypothesis $b^3\neq 1$. The boundary points are

$$D_x=\{(0,1),(0,b^{-1}),(\infty,1),(\infty,b^{-1})\},$$
$$D_y=\{(1,0),(b,0),(1,\infty),(b,\infty)\}.$$

They are distinct and smooth: in the appropriate affine charts the derivative along the finite coordinate is a nonzero multiple of $1+b$ or $1+b^{-1}$. No corner of the square is on the curve. The smooth bidegree-$(2,2)$ curve is geometrically connected: distinct components with zero intersection would all have to be fibers of the same projection, contrary to the total bidegree. It is therefore geometrically irreducible and has genus $(2-1)(2-1)=1$. Its smooth completion descends to K.

The t-power Frobenius on these coordinates is $(x,y)\mapsto(b/y^t,x^t)$. A rational point satisfies $y=x^t$, which excludes all eight boundary points. Hence $C_b(K)=X_b(K)$, with the stated interpretation in L. The final bound is the Hasse–Weil bound. □

3.2 Character components and the constant exception

We now retain all three varying character components on a norm fiber. For $a\in X_b(K)$ put

$$w=(1+a)^B\in U,\qquad v=a^A\in V,\qquad v_+=(1+a)^A\in V. \tag{3.2}$$

Lemma 3.3 (Character decomposition)**.** *The map $C_d\longrightarrow U\times V$, $z\mapsto(z^B,z^A)$, is an isomorphism. Every pair $\xi,\zeta\in\widehat{C_d}$ has a unique decomposition*

$$\xi(z)=\alpha(z^B)\chi(z^A),\qquad \zeta(z)=\eta(z^B)\psi(z^A),\quad \alpha,\eta\in\widehat{U},\quad \chi,\psi\in\widehat{V}.$$

Define the two-character sum and the norm-character exception set by

$$\begin{aligned} S_b(\xi,\zeta)&=\sum_{a\in X_b(K)}\xi(a)\zeta(1+a),\\ \mathcal{E}_{\mathrm{norm}}&=\{(\alpha\circ(z\mapsto z^B),1):\alpha\in\widehat{U}\}. \end{aligned} \tag{3.3}$$

For every $b\in U$ the sum is

$$S_b(\xi,\zeta)=\alpha(b)\sum_{a\in X_b(K)}\eta(w)\chi(v)\psi(v_+).$$

Moreover $(\xi,\zeta)\in\mathcal{E}_{\mathrm{norm}}$ if and only if $\eta=\chi=\psi=1$.

Proof. The two powers have orders dividing A and B, respectively. An element in the kernel has both its A-th and B-th powers equal to one, hence is one because $\gcd(A,B)=1$. The source and target have the same order AB, so the map is an isomorphism. Taking characters gives the unique decomposition. On the fiber $a^B=b$; substitution gives the identity for the sum. Uniqueness also proves the stated equivalence for the exception set. □

3.3 The mixed-character estimate

The required cancellation is supplied by the following proposition. Its proof keeps the descended Lang equations and their Frobenius action explicit: a geometric Kummer equation alone would not identify the finite-field character trace. There are three steps: identify the descended character traces, use boundary inertia to rule out geometric constant components, and apply the tame conductor bound. This keeps the hypotheses of the classical estimate separate from the cyclic-group sieve.

Proposition 3.4 (Mixed characters on a norm fiber)**.** *Let $b\in U$ satisfy $b^3\neq1$. For characters $\eta\in\widehat{U}$ and $\chi,\psi\in\widehat{V}$, not all trivial,*

$$\left|\sum_{a\in X_b(K)}\eta(w)\chi(v)\psi(v_+)\right|\le 8\sqrt{t}. \tag{3.4}$$

The product Lang cover used for this sum is geometrically connected, and hence its smooth projective normalization is geometrically irreducible. Every character quotient is tame, with conductor degree at most eight.

Proof. Let $C_b^\circ = C_b \setminus (D_x \cup D_y)$, with the eight points of Lemma 3.2. This open curve has $C_b^\circ(K) = X_b(K)$. Write $\sigma \in \mathrm{Gal}(L/K)$ for $z \mapsto z^t$, extended as $\sigma \otimes 1$ on every $L \otimes_K R$. The algebraic map

$$\nu(z) = z\,\sigma(z)$$

sends the norm-one torus $\ker N_{L/K}$ to $\mathcal{T} = \mathrm{Res}_{E/K}(\ker N_{L/E})$:

$$N_{L/E}(\nu(z)) = N_{L/E}(z)\,\sigma(N_{L/E}(z)) = N_{L/K}(z) = 1.$$

Its value on K-points is z^A. This construction is needed for the varying norm of $1+a$: exponentiation by A on geometric points would give relative norm w^A, which is not the constant function one.

Set $w_0 = (x+1)(x+b)/x$. The split coordinates of the three K-defined maps to the norm tori are

	first coordinate	second coordinate
$N_{L/E}(1+a)$	w_0	w_0^{-1}
$\nu(a)$	xy	by/x
$\nu(1+a)$	$(x+1)(y+1)$	$(y+1)(x+b)/x$

The first row is a map to $\mathcal{U} = \ker N_{E/K}$, and $\mathcal{U}(K) = U$, $\mathcal{T}(K) = V$. On $\mathcal{U}$ the Frobenius is $z \mapsto z^{-t}$; its Lang equation is $z^{-A} = w_0$. On $\mathcal{T}$ the Frobenius is $(z_0, z_1) \mapsto (z_1^{-t}, z_0^t)$; the Lang equations $z_1^{-t}/z_0 = r_0$, $z_0^t/z_1 = r_1$ give $z_0^B = r_1^t/r_0$. Consequently the product pullback is, over $\overline{K}$,

$$z_U^A = w_0^{-1}, \qquad z_1^B = g_1, \qquad z_2^B = g_2, \tag{3.5}$$

where

$$g_1 = b^{-1}\frac{y^{t-1}}{x^A}, \qquad g_2 = w_0^t\frac{(y+1)^{t-1}}{(x+1)^A}. \tag{3.6}$$

For the second formula, substitute $x + b = xw_0/(x+1)$ in r_1^t/r_0.

These geometric equations inherit their K-structure from the Lang pullbacks. More explicitly, above $F_{C_b}(x,y) = (b/y^t, x^t)$ their Frobenius sends the three displayed covering coordinates to

$$(w_0 z_U,\ xyz_1,\ (x+1)(y+1)z_2). \tag{3.7}$$

This follows from $F(z) = rz$ in the Lang equations. It is compatible with (3.5), since

$$w_0(F_{C_b}) = w_0^{-t}, \quad g_1(F_{C_b}) = (xy)^B g_1, \quad g_2(F_{C_b}) = ((x+1)(y+1))^B g_2.$$

At a rational point $(x,y) = (a, a^t)$, the three deck multipliers in (3.7) are precisely w, v, v_+. Thus the character component has trace $\eta(w)\chi(v)\psi(v_+)$; the inverse Frobenius convention simultaneously inverts the characters and gives the same bound.

We check geometric connectedness and the conductor directly. The divisors of w_0, g_1, g_2 are

supported on the eight boundary points. Their valuations are

P	$\mathrm{ord}_P(w_0)$	$\mathrm{ord}_P(g_1)$	$\mathrm{ord}_P(g_2)$
$(0,1)$	-1	$-A$	-1
$(0,b^{-1})$	-1	$-A$	$-t$
$(\infty,1)$	-1	A	t
(∞,b^{-1})	-1	A	1
$(1,0)$	1	$t-1$	-1
$(b,0)$	1	$t-1$	t
$(1,\infty)$	1	$-(t-1)$	$-t$
(b,∞)	1	$-(t-1)$	1

(3.8)

Indeed x, y have simple zeros and poles on D_x, D_y; $x+1$ has simple zeros at $(1,0),(1,\infty)$ and simple poles at the two points with $x=\infty$, and the corresponding assertion holds for $y+1$. Substitution in (3.6) gives each entry of the table.

For every $m \mid B$, a relation $g_1^r g_2^s = f^m$ in $\overline{K}(C_b)^\times$ would imply, at $(0,1)$ and $(0,b^{-1})$,

$$m \mid Ar+s, \qquad m \mid Ar+ts.$$

Now $\gcd(B,t-1)=\gcd(t^2+1,t-1)=1$ and $\gcd(A,B)=1$. Subtracting forces $m \mid s$, and then $m \mid r$. The valuation -1 of w_0 likewise excludes every nontrivial Kummer power relation for an order dividing A. One can verify simultaneous independence in a common exponent:

$$w_0^{Be_0} g_1^{Ae_1} g_2^{Ae_2} \in \overline{K}(C_b)^{\times AB} \quad \Longrightarrow \quad A \mid e_0,\ B \mid e_1,\ B \mid e_2.$$

Reduction of the first two valuations modulo A gives $A \mid e_0$; their difference modulo B gives $B \mid e_2$, and either valuation then gives $B \mid e_1$. This is independence of the relevant functions modulo powers in $\overline{K}(C_b)^\times$, verified by their local valuations.

At $P_1=(0,1)$ and $P_2=(0,b^{-1})$, the geometric inertia elements, written in $G=\mathbb{Z}/A\mathbb{Z}\times\mathbb{Z}/B\mathbb{Z}\times\mathbb{Z}/B\mathbb{Z}$, are

$$v_1=(-1,-A,-1), \qquad v_2=(-1,-A,-t).$$

Their difference is $(0,0,t-1)$, which generates the third factor because $\gcd(t-1,B)=1$. Combining a suitable multiple of this element with v_1 gives $(-1,-A,0)$. Its order is AB: the first coordinate has order A, the second has order B since $\gcd(A,B)=1$. Thus v_1, v_2 generate all of G. Equivalently, no nontrivial character annihilates both inertia elements. The subgroup generated by geometric inertia is therefore the full deck group, proving geometric connectedness of the product cover and of every character quotient.

All covering degrees are odd, so wild inertia is trivial. For a rank-one character its local Artin conductor is one if the inertia character is nontrivial, and zero otherwise. Thus its conductor degree is the number r of active rows of (3.8), and $r \le 8$. Cancellation can occur at individual rows; we do not presume ramification at every boundary point. For a nontrivial character triple the inertia test proves geometric nontriviality and $r>0$. The genus-one Artin-character Weil bound now gives $(2\cdot 1-2+r)\sqrt{t} \le 8\sqrt{t}$; see [5, §4, Lemma 1] and the Lang construction in [4, pp. 198–199]. In cohomological terms, extend over the inactive boundary points. On the complement of the r active points, H_c^0, H_c^2 vanish and the tame Euler-characteristic formula gives $\dim H_c^1 = 2g-2+r = r$. The eigenvalues have absolute value at most $\sqrt{t}$, by the Weil theorem on the character component of the finite cover. No active boundary point is K-rational, so this trace sum is exactly the sum over $X_b(K)$, with no rational boundary correction. □

Corollary 3.5 (The precise two-character exception)**.** *Let $b \in U$ satisfy $b^3 \neq 1$ and let $\xi, \zeta \in \widehat{C_d}$. Use the character decomposition lemma and* (3.3)*. If $(\xi, \zeta) \in \mathcal{E}_{\mathrm{norm}}$, then*

$$S_b(\xi, \zeta) = \alpha(b) \# X_b(K).$$

In every other case $|S_b(\xi, \zeta)| \leq 8\sqrt{t}$. In particular, the assertion for every nontrivial two-character pair is false: a nontrivial $\alpha \circ N_{L/E}$ with $\zeta = 1$ is constant on the norm fiber.

Proof. The decomposition lemma identifies the exception set with the trivial varying triple. That triple gives the exact constant sum. Every pair outside the set has a nontrivial varying triple, so Proposition 3.4 gives the bound. □

4 Joint generation and double full-order densities

4.1 Two cyclic-group indicators

The mixed-character estimate separates the geometric input from the cyclic-group sieve. Joint generation excludes simultaneous membership in a maximal subgroup; individual full order excludes membership for each element separately. Their local factors and therefore their final densities differ.

Lemma 4.1 (The primitive-element indicator)**.** *For a cyclic group G of order n, put $\theta(n) = \varphi(n)/n$ and $W(n) = 2^{\omega(n)}$. For $z \in G$,*

$$\mathbf{1}_{\mathrm{ord}(z)=n} = \theta(n) \sum_{m \mid n} \frac{\mu(m)}{\varphi(m)} \sum_{\mathrm{ord}(\chi)=m} \chi(z).$$

The constant coefficient is $\theta(n)$, and the total absolute coefficient sum is $\theta(n)W(n)$.

Proof. The character sum factors over the primes $p \mid n$. Its p-factor is $1 - (p-1)^{-1} \sum_{\mathrm{ord}(\chi)=p} \chi(z)$, which is zero on G^p and $p/(p-1)$ off G^p. Multiplication by the local factor $(p-1)/p$ of $\theta(n)$ therefore gives exactly the full-order indicator. Distinct squarefree orders have disjoint character supports; each squarefree divisor contributes absolute coefficient mass one before multiplication by $\theta(n)$. There are $W(n)$ of them. □

Lemma 4.2 (Local averaging in a cyclic group)**.** *Put*

$$\theta_2(n) = \prod_{p \mid n} (1 - p^{-2}), \qquad J_b = \{p \mid A : b \in U^p\},$$
$$\lambda_A(b) = \prod_{p \in J_b} (1 - p^{-1}), \quad H(A) = \prod_{p \mid A} \frac{p+2}{p+1}.$$

The following are exact averages over all A elements of U:

$$\begin{aligned} \frac{1}{A} \sum_{b \in U} \lambda_A(b) &= \theta_2(A), \\ \frac{1}{A} \sum_{b \in U} \lambda_A(b) 2^{|J_b|} &= \theta_2(A) H(A). \end{aligned} \tag{4.1}$$

In particular the average of $\lambda_A(b)\theta_2(B)$ is $\theta_2(A)\theta_2(B) = \theta_2(d)$.

Proof. Fix a generator u and write $b = u^j$, $j \in \mathbb{Z}/A\mathbb{Z}$. For $p \mid A$ the condition $b \in U^p$ is equivalent to $p \mid j$. Reduction of uniform j modulo $\mathrm{rad}(A)$ is uniform; the Chinese remainder theorem makes the residues modulo the distinct primes independent. The probability of $p \in J_b$ is therefore $1/p$. The local expectation for λ_A is

$$(1 - 1/p) + \frac{1}{p}(1 - 1/p) = 1 - p^{-2}.$$

For $\lambda_A(b)2^{|J_b|}$ it is

$$(1 - 1/p) + \frac{2}{p}(1 - 1/p) = (1 - p^{-2})\frac{p+2}{p+1}.$$

Multiplication of these independent local expectations proves both identities. Since $\gcd(A, B) = 1$, their prime supports are disjoint, which proves the last equality. □

4.2 Joint generation

Theorem 4.3 (Joint generation density at the inner boundary)**.** *Use $\theta_2, \lambda_A, J_b, H(A)$ from the local averaging lemma, and put $W(n) = 2^{\omega(n)}$, $r_0 = \gcd(3, A) \le 3$. Let $\mathcal{R}_{d,b} = \mathcal{R}_d \cap X_b(K)$. For every $b \in U$ with $b^3 \neq 1$,*

$$\big||\mathcal{R}_{d,b}| - \lambda_A(b)\theta_2(B)A\big| \le \lambda_A(b)\theta_2(B)\big(8\,2^{|J_b|}W(B) - 6\big)\sqrt{t}. \tag{4.2}$$

For the full family,

$$\big||\mathcal{R}_d| - \theta_2(d)A^2\big| \le \theta_2(d)A\big(8H(A)W(B) - 6\big)\sqrt{t} + 2r_0A. \tag{4.3}$$

Since $H(A) \le W(A)$, this implies

$$|\mathcal{R}_d| = \theta_2(d)t^2 + O(t^{3/2}W(d)), \qquad |\mathcal{R}_d| \sim \theta_2(d)t^2.$$

The fixed-fiber population is $O(t)$, and its coefficient depends on b. If b is primitive in U, then J_b is empty and the coefficient is $\theta_2(B)$. Moreover $|T_d - A^2| \le 2A\sqrt{t} + r_0A$, so the actual proportion of jointly generating slopes among the ordinary intersection is $|\mathcal{R}_d|/T_d = \theta_2(d) + O(W(d)/\sqrt{t})$.

Proof. The group isomorphism gives

$$\langle a, 1 + a\rangle = C_d \quad \Longleftrightarrow \quad \langle b, w\rangle = U \text{ and } \langle v, v_+\rangle = V.$$

For $p \mid A$ with $b \notin U^p$ the first condition is automatic. For $p \in J_b$ it requires $w \notin U^p$. By orthogonality its indicator is the product of the factors

$$1 - \frac{1}{p}\sum_{\eta^p=1} \eta(w), \qquad p \in J_b.$$

Each factor has constant coefficient $1 - 1/p$ and total absolute coefficient sum $2(1 - 1/p)$. The second condition has indicator

$$\prod_{p \mid B}\left(1 - \frac{1}{p^2}\sum_{\chi^p=\psi^p=1} \chi(v)\psi(v_+)\right).$$

Here each factor has constant coefficient $1 - 1/p^2$ and total absolute coefficient sum $2(1 - 1/p^2)$. Characters belonging to different prime-primary components have unique products, so multiplying

these expansions gives constant coefficient $c_b = \lambda_A(b)\theta_2(B)$ and total absolute coefficient sum $c_b 2^{|J_b|}W(B)$. The Hasse error for the trivial triple is $2c_b\sqrt{t}$; Proposition 3.4 bounds all other terms by $8\sqrt{t}$. This proves (4.2).

The local averaging lemma supplies (4.1). Multiplication by $\theta_2(B)$ turns $\theta_2(A)$ into $\theta_2(d)$; it also turns the weighted average into $\theta_2(d)H(A)$. These identities average over all $b \in U$, before separating the exceptional fibers.

There are precisely r_0 exceptional b with $b^3 = 1$. For every b, each $w \in U$ allows at most two roots of $X^2 + (1 + b + w)X + b$. Thus $\#X_b(K) \le 2A$ even in an exceptional fiber. Its contribution differs from the formal main term $c_b A \in [0, A]$ by at most $2A$. Sum (4.2) over the good fibers, bound the positive error coefficients by their sum over all b, and use the two exact averages. This gives (4.3). It includes all exceptional and subfield points, rather than applying the smooth-fiber bound to them. Finally $W(d) = d^{o(1)} = t^{o(1)}$, while $\theta_2(d) \ge 1/2$: the latter follows by comparison with the telescoping product $\prod_{n=2}^{\infty}(1 - n^{-2}) = 1/2$. Replacing $A^2 = (t + 1)^2$ by t^2 gives the stated asymptotic. For T_d sum the Hasse bound over the good fibers and use $|\#X_b(K) - A| \le A$ on each exceptional fiber. This proves its bound and, by division, the density assertion. □

4.3 Both linked slopes of full order

Theorem 4.4 (Double full-order statistics and all-dimensional existence)**.** *Define* $\theta(n) = \varphi(n)/n$ *and*

$$D_{d,b} = \#\{a \in X_b(K) : \operatorname{ord}(a) = \operatorname{ord}(1 + a) = d\}, \qquad D_d = \sum_{b \in U} D_{d,b}.$$

If b *is not primitive in* U*, then* $D_{d,b} = 0$*. For primitive* b*, put* $c = \theta(A)\theta(B)^2$*. Then*

$$|D_{d,b} - cA| \le c\big(8W(A)W(B)^2 - 6\big)\sqrt{t}. \tag{4.4}$$

In particular,

$$|D_d - \theta(d)^2A^2| \le \theta(d)^2A\big(8W(A)W(B)^2 - 6\big)\sqrt{t}, \tag{4.5}$$

and $D_d \sim \theta(d)^2t^2$ *as* $s \to \infty$*. Define the explicit analytic range*

$$\mathcal{S}_{\mathrm{an}} = \{s \in \mathbb{Z}_{\ge 2} : A > (8W(A)W(B)^2 - 6)\sqrt{t}\}, \tag{4.6}$$

with $t = 2^s$*,* $A = t + 1$*, and* $B = t^2 + 1$*. For every* $s \in \mathcal{S}_{\mathrm{an}}$ *and every primitive* $b \in U$*,* $D_{d,b} > 0$*. This set contains all sufficiently large integers* s*. Moreover,* $D_d > 0$ *for every integer* $s \ge 2$*.*

Proof. Order d forces $a^B = b$ to have order A, which proves the vanishing assertion. For primitive b, the two full-order conditions are exactly that w is primitive in U and v, v_+ are primitive in V. Apply Lemma 4.1 separately to these three variables. For a cyclic group of order n it has constant coefficient $\theta(n)$ and total absolute coefficient sum $\theta(n)W(n)$. The product therefore has constant coefficient c and total absolute coefficient sum $cW(A)W(B)^2$. The trivial triple contributes $c\#X_b(K)$, with Hasse error $2c\sqrt{t}$; every other triple has the mixed bound $8\sqrt{t}$. This proves (4.4). Sum over the $\varphi(A) = A\theta(A)$ primitive norm fibers and use $\theta(d) = \theta(A)\theta(B)$ to obtain (4.5). The relative error tends to zero because $W(A)W(B)^2 = t^{o(1)}$. The stated inequality makes the lower bound positive in every primitive norm fiber. For all-dimensional existence, Proposition S8.1 below gives $D_d \ge \Delta_t > 0$ for $s \ge 33$, independently of the preceding three-character sieve. The exact certificates in Appendix A prove $D_d > 0$ for each $2 \le s \le 32$. □

5 Uniform relative traces and the exceptional trace groups

The two full-order conditions become equivalent when the linked elements are Frobenius conjugates. We first allow their difference to be any nonzero $c \in E$, obtaining a distribution theorem on the whole relative-trace field. We classify its two exceptional trace groups. Their intersections with K^* are both $\{1\}$, recovering the two exceptional trace-one parameters. The case $c = 1$ is a subfamily of D_d. We then describe the distribution within each trace-norm class and transfer the classification to every nonzero K-norm fiber.

5.1 A uniform four-point character estimate

For $c \in E$ put

$$Q_c = \#\{a \in C_d : a + a^{t^2} = c\}, \qquad M_c = \#\{a \in C_d : \operatorname{ord}(a) = d,\ a + a^{t^2} = c\}.$$

The notation M_c agrees with the projective-to-affine comparison in Supplementary Section S7. In particular $M_1 = \Delta_t$ for Proposition S8.1.

Theorem 5.1 (Uniform nonzero relative-trace slices)**.** *Let $t = 2^s$, $s \geq 2$, and $c \in E^*$. The map*

$$a = \frac{c}{1+\gamma}, \qquad \tau = \gamma + \gamma^{-1}, \qquad w = \tau/c^2, \tag{5.1}$$

is a bijection from $\{\gamma \in V : w \in U\}$ onto the set counted by Q_c. It restricts to a bijection onto the set counted by M_c when $\operatorname{ord}(\gamma) = B$ and $\operatorname{ord}(w) = A$. For every nontrivial character pair (η, χ) of $U \times V$,

$$\left| \sum_{\substack{\gamma \in V \\ (\gamma+\gamma^{-1})/c^2 \in U}} \eta((\gamma+\gamma^{-1})/c^2)\chi(\gamma) \right| \leq 4\sqrt{t}. \tag{5.2}$$

Consequently, uniformly in $c \in E^*$,

$$|M_c - \theta(d)Q_c| \leq 4\theta(d)(W(d)-1)\sqrt{t}, \tag{5.3}$$

$$|Q_c - (t+1)| \leq 2\sqrt{t}, \tag{5.4}$$

$$|M_c - \theta(d)(t+1)| \leq \theta(d)(4W(d)-2)\sqrt{t}. \tag{5.5}$$

The zero slice contains no element of order d: $M_0 = 0$.

Proof. If $\gamma \in V$ and $w \in U$, then $\gamma \neq 1$ and

$$a^{t^2} = a + c, \qquad a^B = c^2/\tau = w^{-1}, \qquad a^{t^2-1} = \gamma.$$

Thus $a \in C_d$. Conversely an element counted by Q_c has $\gamma = (a+c)/a \in V$ and satisfies the same identities, so the maps are inverse. Since $\gcd(A, B) = 1$ and $\gcd(B, t^2-1) = 1$, the displayed powers recover the A-part and B-part of the order of a. Thus a has order d exactly when w and γ have orders A and B. If $c = 0$, then $a \in E$ and its order divides $\gcd(d, t^2-1) = A < d$, proving the final assertion.

Every $c \in E^*$ has a unique decomposition $c = c_0 u$ with $c_0 \in K^*$ and $u \in U$: take $c_0^2 = N_{E/K}(c)$ and $u = c/c_0$. Writing $w_0 = \tau/c_0^2$, we have $w = u^{-2}w_0$. The two summation sets are identical, and their character sums satisfy

$$S_c(\eta, \chi) = \eta(u^{-2})S_{c_0}(\eta, \chi), \qquad Q_c = Q_{c_0}.$$

The extra factor has absolute value one. It therefore suffices to establish the character estimate for $c \in K^*$, which we assume during the following geometric argument. The primitive indicator will still be applied to the original $w = u^{-2}w_0$; the full-order populations M_c need not equal M_{c_0}.

Over $\overline{K}$, compactify the parameter curve in $\mathbb{P}^1 \times \mathbb{P}^1$ as

$$Y_c : \quad (x+1)^2(y+1)^2 = c^4xy, \qquad F_Y(x,y) = (y^{-t}, x^t). \tag{5.6}$$

The displayed Frobenius is the inherited torus descent; it is not coordinatewise Frobenius on a split model. Its open K-points have $x = \gamma$, $y = \gamma^t$, $\gamma \in V$, and $N_{E/K}(\tau) = c^4$. Hence they are precisely the summation set. The affine partial derivatives are c^4y, c^4x; their only common zero is off the curve. The boundary points are $P_0 = (0,1)$, $P_\infty = (\infty, 1)$, $Q_0 = (1,0)$, $Q_\infty = (1,\infty)$. Writing $u = y + 1$ at P_0 gives $(1+x)^2u^2 = c^4x(1+u)$, so the derivative with respect to x is $c^4 \neq 0$ and $v_{P_0}(x) = 2$. Reciprocal and coordinate symmetries give smoothness at the other three points. Thus the bidegree-$(2,2)$ curve is smooth. It is geometrically connected: distinct smooth components would be disjoint, whereas any decomposition of the bidegree $(2,2)$ into nonzero effective bidegrees contains two components with positive intersection. It is therefore geometrically irreducible, and adjunction gives genus one. Frobenius cycles

$$P_0 \longmapsto Q_0 \longmapsto P_\infty \longmapsto Q_\infty \longmapsto P_0.$$

The boundary has no K-point, and Hasse–Weil proves (5.4).

To realize the character sum by the actual Lang covers, set

$$w = \frac{(x+1)^2}{c^2x}, \qquad h = \frac{y^t}{x}.$$

The two split coordinates of the map to $\ker N_{E/K}$ are (w, w^{-1}). As in the Lang construction [4], the pullbacks from $\ker N_{E/K}$ and $\mathrm{Res}_{E/K}(\ker N_{L/E})$ are

$$z_U^A = w^{-1}, \qquad z_V^B = h.$$

Indeed their split Lang equations are respectively $z_U^{-t}/z_U = w$ and $z_1^{-t}/z_0 = x$, $z_0^t/z_1 = y$. The covering Frobenius sends (z_U, z_V) to (wz_U, xz_V) over F_Y; the identities $w(F_Y) = w^{-t}$ and $h(F_Y) = x^Bh$ verify compatibility. Thus its deck multipliers at a K-point are w and γ, and the character trace is $\eta(w)\chi(\gamma)$, with the inverse convention merely inverting both characters.

Multiplication by the nonzero constant c^{-2} changes no boundary valuation. At Q_0, the local equation gives $v(y) = 2$ and $v(x+1) = 1$; inversion and coordinate interchange give the other rows. The valuations are

	w	h
P_0	-2	-2
P_∞	-2	2
Q_0	2	$2t$
Q_∞	2	$-2t$

(5.7)

For $0 \le e_0 < A$, $0 \le e_1 < B$, an ABth-power relation $(w^{-1})^{Be_0}h^{Ae_1} \in \overline{K}(Y_c)^{\times AB}$ would have valuation $2Be_0 - 2Ae_1$ at P_0. Reduction modulo A and B forces $e_0 = e_1 = 0$. The product cover is therefore geometrically connected with full group $U \times V$. Every nontrivial character pair is ramified at all four boundary points: the coordinate valuations generate each odd cyclic factor, and the two character values cannot cancel because their orders divide the coprime integers A, B. The ramification is tame, with local conductor one. The associated rank-one finite-monodromy sheaf is

geometrically nontrivial and pure of weight zero. Its H_c^0, H_c^2 vanish; Grothendieck–Ogg–Shafarevich gives $\dim H_c^1 = 2g - 2 + 4 = 4$. The classical curve-character estimate [5, Lemma 1] now gives (5.2).

The phase identity now gives the same character bound for every $c \in E^*$. The product of the two primitive indicators, applied to the original pair (w, γ), has constant coefficient $\theta(A)\theta(B) = \theta(d)$ and total absolute coefficient mass $\theta(d)W(d)$. The nonconstant terms give (5.3). Adding the Hasse error of the constant term gives (5.5). The identity $Q_c = Q_{c_0}$ transfers the unfiltered estimate as well. All constants are therefore uniform on E^*. □

The reciprocal-trace substitution is classical [17, §4]. After the family-specific smoothness, connectedness and ramification checks above, the $4\sqrt{t}$ character bound is a direct application of the classical curve-character theorem; the quantitative conclusion is its uniform specialization to these exact-order slices. Only $c = 1$ makes $a + c = 1 + a$, so the other slices describe relative-trace distributions.

The counting object is classical: it occurs in the prescribed-order and prescribed-trace frameworks of [22, 23, 10], and in the line framework of [6]. Appendix D gives exact specializations of these results and of [25, 26, 27, 21]. Their displayed sufficient conditions do not establish the present uniform estimate or complete classification; the $s = 2$ positivity case is already covered by [25, Theorem 1.6]. The appendix also distinguishes the primitive-normal aggregate in [11] from a fixed relative-trace population.

5.2 The exact unfiltered center

The center Q_c can be written as a classical binary Kloosterman sum. This separates the fluctuation of the whole slice from the error caused by the order restrictions. Put $\psi(z) = (-1)^{\mathrm{Tr}_{K/\mathbb{F}_2}(z)}$ and $\mathrm{Kl}_t(\lambda) = \sum_{z \in K^*} \psi(z + \lambda/z)$.

Lemma 5.2 (Unfiltered trace slices)**.** *For every $c \in E^*$,*

$$Q_c = t + 1 + \mathrm{Kl}_t(N_{E/K}(c)^{-2}). \tag{5.8}$$

Proof. First assume $c \in K^*$. For $b \in U$, the polynomial $X^2 + cX + b$ gives two roots with relative trace c exactly when it is irreducible over E. The binary quadratic criterion therefore gives

$$Q_c = A - \sum_{b \in U} \psi\big((b + b^{-1})/c^2\big).$$

For every $r \in K$, the total number of roots of $X^2 + rX + 1$ in the two groups K^* and U is two. For $r \neq 0$ the polynomial splits in one of the two tori; at $r = 0$, its root 1 is counted once in each. Hence, for $\alpha \neq 0$, the two torus character sums add to $2\sum_{r \in K} \psi(\alpha r) = 0$. The sum over K^* equals $\mathrm{Kl}_t(\alpha^2)$ after $z \mapsto \alpha z$. Taking $\alpha = c^{-2}$ gives $Q_c = t + 1 + \mathrm{Kl}_t(c^{-4})$. For general $c = hu$, the preceding proof gives $Q_c = Q_h$ and $h^{-4} = N_{E/K}(c)^{-2}$, proving the stated identity. □

5.3 A prime sieve and complete trace classification

The following is the usual prime-sieve inequality for freeness [22, eq. (3.2)]; see also [7, 13]. Here it is applied after the four-point estimate. We include the calculation to specify its constant.

Lemma 5.3 (Prime sieve on a nonzero trace slice)**.** *Let $e \mid \mathrm{rad}(d)$ be a squarefree core and let $p_1, \dots, p_r$ be the primes dividing d but not e. Put*

$$\delta = 1 - \sum_{i=1}^{r} \frac{1}{p_i} > 0, \qquad \Lambda = 2 + \frac{r - 1}{\delta}.$$

For $r = 0$ these definitions give $\delta = \Lambda = 1$. For every $c \in E^*$,

$$M_c \geq \theta(e)\delta\left[t + 1 - (4W(e)\Lambda - 2)\sqrt{t}\right]. \tag{5.9}$$

Proof. Identify $U \times V$ with a cyclic group of order d and let $N_e(c)$ count those (w, γ) on Y_c avoiding its pth-power subgroup for every $p \mid e$. The primitive indicator with the primes of e gives

$$|N_e(c) - \theta(e)A| \leq \theta(e)(4W(e) - 2)\sqrt{t}.$$

For $p \nmid e$, the indicator for ep minus $(1 - p^{-1})$ times the indicator for e contains only nontrivial character pairs, of total absolute coefficient mass $\theta(e)(1 - p^{-1})W(e)$. Thus

$$|N_{ep}(c) - (1 - p^{-1})N_e(c)| \leq 4\theta(e)(1 - p^{-1})W(e)\sqrt{t}.$$

The pointwise union bound for the omitted-prime failures gives

$$M_c \geq \sum_{i=1}^{r} N_{ep_i}(c) - (r - 1)N_e(c).$$

Writing the right side as $\delta N_e(c)$ plus the r differences just estimated, and using $\sum_i (1 - p_i^{-1}) = r - 1 + \delta$, proves the bound. For $r = 0$ it reduces to the first displayed estimate. □

To pass from a representative trace $h \in K^*$ to its whole trace-norm class, we use the reciprocal-trace supports. For $h \in K^*$ define

$$T_h = \{(\gamma + \gamma^{-1})/h^2 \in U : \gamma \in V,\ \mathrm{ord}(\gamma) = B\}, \qquad n_h = 2|T_h|.$$

The factor two records the pair γ, γ^{-1} over each reciprocal trace. The class hU consists of exactly those $c \in E^*$ with $N_{E/K}(c) = h^2$. They satisfy $T_h = T_h^{-1}$: raising a primitive γ to the tth power preserves its order, fixes h, and sends $(\gamma + \gamma^{-1})/h^2$ to its inverse in U. This Frobenius symmetry is essential to the following reduction. The underlying Chinese remainder description is classical in unitary Cayley graphs; see [28, §1]. We give the elementary argument needed here.

Lemma (Inverse supports with at most two prime divisors)**.** *Let G be cyclic of odd order $n > 1$, let $T = T^{-1} \subseteq G$, and put $\rho = \mathrm{rad}(n)$. Let H have order n/ρ, let S be the image of T in G/H, and define*

$$F_T(u) = 2\#\{z \in T : \mathrm{ord}(zu^{-2}) = n\}.$$

If $\rho = p$ is prime, the zero set of F_T is G when $T = \varnothing$, is G^p when $S = \{1\}$, and is empty otherwise.

Suppose $\rho = pq$ for distinct primes and $F_T(1) > 0$. Then $F_T(u) > 0$ for every $u \in G$, unless $S = \{v, v^{-1}\}$ for a generator v of G/H. In that exceptional case write $v = (a, b)$ under $G/H \cong C_p \times C_q$. The zeros are exactly the u with

$$\overline{u}^2 \in \{(a, b^{-1}), (a^{-1}, b)\},$$

and there are $2n/(pq)$ of them. In particular $|S| \geq 3$ excludes this case; the sufficient condition $|T| > 2n/\rho$ also does so.

Finally, if $\rho = pq$, $S \subseteq \{1\} \times C_q$ and $|S| \geq 2$, the zero set is exactly G^p.

Proof. An element generates G exactly when its image has nonidentity coordinate at every prime dividing ρ. Hence

$$F_T(u) = 0 \quad \Longleftrightarrow \quad T \subseteq \bigcup_{p|\rho} u^2 G^p.$$

For $\rho = p$, a nonempty S can be covered by one coordinate value only when it is a singleton. Inversion then forces that value to be 1, since p is odd. Squaring is bijective, which proves the first assertion.

For $\rho = pq$ and $F_T(1) > 0$, choose $v = (a, b) \in S$ with $a, b \neq 1$. Inversion puts v^{-1} in S, and oddness gives $a \neq a^{-1}$, $b \neq b^{-1}$. A zero must cover both points by its row and column. Its squared image must therefore be one of the two mixed corners (a, b^{-1}), (a^{-1}, b). The zero set is inversion invariant, so if one mixed corner occurs then both occur. The intersection of their row–column unions is precisely $\{v, v^{-1}\}$. Thus S must be that pair. Conversely that pair has exactly the two stated zero corners. Each lifts to n/ρ elements, proving the formula and the cardinality consequences.

For the final assertion, the first coordinate 1 covers all of S. Any other first coordinate would require its column to cover two distinct points of S, which is impossible. □

By Theorem 5.1, $M_{hu} = F_{T_h}(u)$. Thus, when A has two distinct prime divisors, a primitive support point z_0 and one additional point whose image is outside $\{\overline{z}_0, \overline{z}_0^{-1}\}$ certify positivity throughout hU. The additional condition cannot be dropped: in $G = \langle g \rangle$ of order 15, $T = \{g, g^{-1}\}$ has $F_T(1) = 4$ but $F_T(g^2) = F_T(g^{13}) = 0$. This is a cyclic-group reduction, not an additional geometric estimate. Neither its proof nor the support certificates below uses the classification to be proved.

Theorem 5.4 (Complete nonzero-relative-trace classification)**.** *For $s \geq 2$ and $c \in E^* = \mathbb{F}_{2^{2s}}^*$,*

$$M_c = 0 \quad \Longleftrightarrow \quad (s = 3 \text{ and } c^9 = 1) \quad \text{or} \quad (s = 5 \text{ and } c^{11} = 1). \tag{5.10}$$

Thus the exceptional sets are the subgroup of order 9 at $s = 3$ and the subgroup of order 11 at $s = 5$. The analytic estimate and prime sieve give positivity in every nonzero slice for $s \geq 10$, and also for $s = 8$. On restricting to K^*, each exceptional subgroup meets K^* only at 1. In particular the trace-one classification is

$$\Delta_{2^s} = M_1 > 0 \quad \Longleftrightarrow \quad s \notin \{3, 5\} \qquad (s \geq 2). \tag{5.11}$$

Proof. For $s \geq 33$, the integer bounds in Appendix A give $\sqrt{t} \geq 4W(d)$. Theorem 5.1 then proves $M_c > 0$ for every $c \neq 0$. For $s = 8$ and $10 \leq s \leq 32$, Appendix B supplies a core $e \in \{1, 3\}$ and an integer $C_s \geq 4W(e)\Lambda - 2$ with $(t+1)^2 > tC_s^2$. Lemma 5.3 proves positivity in all these slices.

Let Φ_j denote the reduction modulo two of the jth cyclotomic polynomial and put $H_j(X) = X^{\varphi(j)}\Phi_j(X + X^{-1})$. This classical Q-transform [24, §1] records the exact order of a reciprocal trace. Since B is odd, Φ_B has simple roots, all primitive in V. In particular

$$\Delta_t = \deg \gcd(\Phi_B, H_A) = \sum_{e|\mathrm{rad}(B)} \mu(e) \deg \gcd(X^{B/e} + 1, H_A). \tag{5.12}$$

Appendix B retains the exact small trace-one counts. For the exceptional supports, exact binary polynomial divisions give

$$\begin{aligned} \gcd(H_j, \Phi_{65}) &= 1 \qquad (j = 1, 3, 9), \\ \gcd(H_j, \Phi_{1025}) &= 1 \qquad (j = 1, 3, 33), \\ H_{11} &\mid \Phi_{1025}. \end{aligned}$$

Thus $T_1 = \varnothing$ at $t = 8$, since $1, 3, 9$ are all possible reciprocal-trace orders in U. At $t = 32$, the twenty roots of H_{11} give all ten order-11 reciprocal traces, while the other possible orders $1, 3, 33$ have no primitive preimage. Hence $T_1 = C_{11} \setminus \{1\}$. The inverse-support lemma now gives exactly the empty group $C_9 = U$ in the first case, and $U^3 = C_{11}$ in the second. For the positive-population refinement below, the same divisions give $H_1 = \Phi_3$, $H_3 = \Phi_5$ and

$$\gcd(H_{33}, X^{1025} + 1) = \gcd(H_{33}, X^{205} + 1), \qquad \deg \gcd(H_{33}, X^{205} + 1) = 20,$$

with $\gcd(H_{33}, X^{41} + 1) = \gcd(H_{33}, X^5 + 1) = 1$. These identities also identify the order drop in the latter support obstruction.

For the remaining finite range $s = 2, \ldots, 7, 9$, every $A = t + 1$ has at most two distinct prime divisors. Appendix B supplies 755 primitive support witnesses, one for every $h \in K^*$ except $(s, h) = (3, 1), (5, 1)$. They prove $M_h > 0$ directly through Theorem 5.1. In the 731 classes where A has two distinct prime divisors, a second witness has image outside the inverse pair of the first, so $|\overline{T_h}| \geq 3$. The inverse-support lemma gives $M_{hu} > 0$ for every $u \in U$. The independent verifier checks both kinds of witnesses in degree-$4s$ binary polynomial fields, without an all-E coverage mask. Unique decomposition $c = hu$ proves every remaining slice positive. Together with the analytic range and the two exact support obstructions, this proves (5.10). Restriction to K^*, which meets each exceptional group only at 1, proves (5.11). The earlier 3319 full-trace witness families remain a separate check, rather than a premise of this reduction. □

Every trace in either exceptional group has norm 1. Lemma 5.2 and $Q_c = Q_1$ therefore give $Q_c = 4$ at $s = 3$ and $Q_c = 44$ at $s = 5$ on these groups. The unfiltered slices are nonempty; the exact-order filter causes the vanishing.

5.4 Distribution within a fixed trace-norm class

The preceding classification concerns zeros. A cyclic convolution also describes the positive populations and their fluctuation. Retain T_h and n_h from the inverse-support reduction above.

Proposition 5.5 (Cyclic convolution and a variance bound)**.** *For $h \in K^*$ and $u \in U$,*

$$M_{hu} = 2\#\{z \in T_h : \operatorname{ord}(zu^{-2}) = A\}. \tag{5.13}$$

The mean $\mu_h = A^{-1} \sum_{u \in U} M_{hu}$ and variance satisfy

$$\mu_h = \theta(A) n_h, \tag{5.14}$$

$$\frac{1}{A} \sum_{u \in U} |M_{hu} - \mu_h|^2 \leq 16\theta(A)(1 - \theta(A))\theta(B)^2 W(B)^2 t. \tag{5.15}$$

Moreover $M_{huv} = M_{hu}$ for every v in the subgroup of U of order $A/\operatorname{rad}(A)$. If $n_h > 0$, the number Z_h of empty slices in hU satisfies

$$\frac{Z_h}{A} \leq \frac{16\theta(A)(1 - \theta(A))\theta(B)^2 W(B)^2 t}{\mu_h^2}.$$

Proof. In Theorem 5.1, the admissible primitive γ have $z = (\gamma + \gamma^{-1})/h^2 \in U$. Each z has exactly two preimages, since the quadratic $X^2 + h^2 z X + 1$ has distinct roots γ, γ^{-1}. The remaining order condition is that zu^{-2} generate U, proving (5.13). As $u \mapsto u^2$ permutes U, each z contributes to precisely $\varphi(A)$ traces. This gives the mean.

For a character η of U put

$$C_A(\eta) = \sum_{\mathrm{ord}(w)=A} \overline{\eta(w)}, \qquad S_h(\eta) = 2 \sum_{z \in T_h} \eta(z).$$

The classical Ramanujan-sum identity gives $C_A(\eta) = \mu(j)\varphi(A)/\varphi(j)$ when $j = \mathrm{ord}(\eta)$. Fourier inversion on U yields

$$M_{hu} = \frac{1}{A} \sum_{\eta \in \widehat{U}} C_A(\eta) S_h(\eta) \eta(u^{-2}).$$

For $\eta \neq 1$, expand only the primitive indicator on V. All resulting character pairs are nontrivial, so Theorem 5.1 gives $|S_h(\eta)| \leq 4\theta(B)W(B)\sqrt{t}$. Orthogonality, again using the permutation $u \mapsto u^2$, therefore gives the exact identity

$$\frac{1}{A} \sum_{u} |M_{hu} - \mu_h|^2 = \frac{1}{A^2} \sum_{\eta \neq 1} |C_A(\eta)|^2 |S_h(\eta)|^2.$$

Parseval for the primitive indicator says $\sum_\eta |C_A(\eta)|^2 = A\varphi(A)$. Subtracting its trivial-character term $\varphi(A)^2$ proves (5.15). Only characters of squarefree order contribute. They are trivial on the subgroup of order $A/\operatorname{rad}(A)$, proving the claimed period. Each zero contributes μ_h^2 to the squared deviation, which proves the final bound. □

The Fourier identities are standard. Their use here retains the four-point geometric estimate and removes $W(A)$ from the variance bound. The equality $Q_{hu} = Q_h$ thus coexists with a nonconstant distribution of the exact-order counts.

Corollary 5.6 (The exceptional norm-one profiles)**.** *For $t = 8$, $T_1 = \varnothing$ and $M_c = 0$ for every $c \in U = C_9$. For $t = 32$, $T_1 = C_{11} \setminus \{1\} \subset U = C_{33}$ and*

$$M_c = \begin{cases} 0, & c \in C_{11}, \\ 20, & c \in C_3 \setminus \{1\}, \\ 18, & c \in U \setminus (C_{11} \cup C_3). \end{cases} \tag{5.16}$$

The three populations occur on 11, 2, and 20 traces, respectively. Throughout U, the unfiltered counts are 4 at $t = 8$ and 44 at $t = 32$.

Proof. For $t = 8$ the reciprocal-trace exclusions in the proof of Theorem 5.4 already show $T_1 = \varnothing$. For $t = 32$, let $H_j(X) = X^{\varphi(j)}\Phi_j(X + X^{-1})$ as above. Exact polynomial division gives the following complete table for $\gamma \in V$ and $\tau = \gamma + \gamma^{-1} \in U$:

$\mathrm{ord}(\tau)$	1	3	11	33
$\#\gamma$	0	4	20	20
$\mathrm{ord}(\gamma)$	–	5	1025	205.

The $j = 1, 3, 33$ entries are the polynomial identities already used above. For the remaining entry, H_{11} has degree 20 and divides Φ_{1025} over $\mathbb{F}_2$; the accompanying independent verifier checks the division. Thus all ten order-11 reciprocal traces, and no others in U, arise from primitive γ. This proves $T_1 = C_{11} \setminus \{1\}$.

More generally let $U = C_\ell \times C_p$ for distinct odd primes ℓ, p, and suppose $T_h = C_p \setminus \{1\}$. If $u \in C_p$, the C_ℓ component of zu^{-2} is trivial, so the convolution is zero. If $u \in C_\ell \setminus \{1\}$, all $p - 1$ choices of z give full order. In the remaining case exactly one choice, $z = u_p^2$, has trivial C_p component; the other $p - 2$ choices give full order. Formula (5.13), with $(\ell, p) = (3, 11)$, gives (5.16). The unfiltered counts follow from the table and the $t = 8$ order-3 reciprocal traces. □

5.5 Transfer to arbitrary nonzero norm fibers

The subgroup count has a direct interpretation on every coset of C_d. This transfer uses the coprime factorization $L^* = K^* \times C_d$. Generator counts in cyclic cosets are classical; see, for example, [11, Lemma 1]. The transfer below is a consequence of the preceding theorems, rather than a separate character-sum method.

Corollary 5.7 (Prescribed norm and relative trace)**.** *Let $\nu \in K^*$ have order e, and let $\lambda \in K^*$ be the unique element with $\lambda^4 = \nu$. The greatest order attained on $N_{L/K}^{-1}(\nu)$ is ed. Put*

$$M_{\nu,c} = \#\{a \in L^* : N_{L/K}(a) = \nu,\ a + a^{t^2} = c,\ \mathrm{ord}(a) = ed\}.$$

Then $M_{\nu,0} = 0$ and, for $c \in E^*$,

$$M_{\nu,c} = M_{c/\lambda}, \qquad |M_{\nu,c} - \theta(d)(t+1)| \le \theta(d)(4W(d) - 2)\sqrt{t}. \tag{5.17}$$

In particular,

$$M_{\nu,c} = 0 \quad \Longleftrightarrow \quad (s = 3 \text{ and } c^{36} = \nu^9) \quad \text{or} \quad (s = 5 \text{ and } c^{44} = \nu^{11}). \tag{5.18}$$

The unfiltered population with the same norm and trace is

$$Q_{\nu,c} = t + 1 + \mathrm{Kl}_t(\nu / N_{E/K}(c)^2).$$

If ν is primitive in K^*, these are counts of primitive elements of the ambient field L.

Proof. The integers $t-1$ and d are coprime. Every element of the norm fiber has a unique expression $a = \lambda x$, $x \in C_d$. The fourth-power map permutes K^*, so $\mathrm{ord}(\lambda) = e$ and $\mathrm{ord}(a) = e\,\mathrm{ord}(x)$. Its relative trace is $\lambda(x + x^{t^2})$. The bijection proves the count identity, the maximum-order assertion and the bound. The exceptional sets are therefore λC_9 at $s = 3$ and λC_{11} at $s = 5$. Taking fourth powers gives the displayed equations, with equivalence because fourth powers also permute E^*. Finally $Q_{\nu,c} = Q_{c/\lambda}$ and $N_{E/K}(c/\lambda)^{-2} = \nu / N_{E/K}(c)^2$. Lemma 5.2 gives the final formula. When $e = t - 1$, the maximum order ed equals $t^4 - 1$. □

5.6 Consequences for adjacent pairs and affine groups

For $c = 1$, (5.1) gives $1 + a = a^{t^2}$, so $D_d \ge \Delta_t$. Theorem 5.4 proves $D_d > 0$ except possibly at $s = 3, 5$. At those two parameters the exact double-order certificates in Appendix A give elements outside the trace-one slice; their full-family counts are $24,240$. This proves all-parameter existence using only two exceptional field certificates. The earlier certificates for $2 \le s \le 32$ remain additional reproducible checks.

Every slope counted by D_d lies in no proper subfield of L, because $d > t^2 - 1$. Thus $\mathrm{ord}_d(2) = 4s$, and Corollary 2.5 makes each of these slopes, with any nonzero translation, a realization of $L^+ \rtimes C_d$. For elements counted by M_c, the enlarged trace range distinguishes two normality types over K. Put $T(a) = a^t$; then $(T+1)^2 a = c$ and $(T+1)^3 a = c + c^t$. For $c \in K^*$ the K-order is $(X+1)^3$, so these are 1-normal elements. For $c \in E \setminus K$ it is $(X+1)^4$, so the elements are normal over K; the latter criterion is classical for binary quartic extensions [11, §3]. The 1-normal terminology follows [21]. The same estimate thus treats both normality types through their relative traces. Their $O(t)$ slice populations and the $O(t^2)$ adjacent-pair populations describe different parts of the norm geometry.

6 General index bounds and the norm obstruction

We return to $q = 2^n$, $d \mid q-1$ and $k = (q-1)/d$. The quartic result treats the special index $q^{1/4}-1$. For comparison, this section gives an elementary sufficient range for all admissible indices and a structural obstruction just beyond the quarter-power scale.

6.1 Jacobi bounds and fixed indices

Proposition 6.1 (A positive lower bound for realizing slopes)**.** *Let $q = 2^n$, let $d > 1$ divide $q-1$, and set $k = (q-1)/d$. Write $\omega(d)$ for the number of distinct prime divisors of d, and put*

$$b_k = \frac{(k-1)(k-2)}{k^2}.$$

For every divisor e of $q-1$, with $\ell = (q-1)/e$, one has

$$\left| T_e - \frac{q+1-3\ell}{\ell^2} \right| \le b_\ell \sqrt{q}. \tag{6.1}$$

Consequently,

$$|\mathcal{R}_d| \ge \frac{q}{k^2}\left(1 - \sum_{p \mid d} \frac{1}{p^2}\right) - \frac{3k-1}{k^2} - (\omega(d) + b_k)\sqrt{q}, \tag{6.2}$$

where the sum runs over the distinct prime divisors of d. In particular,

$$|\mathcal{R}_d| > \frac{3q}{4k^2} - \frac{3k-1}{k^2} - (\omega(d) + b_k)\sqrt{q}. \tag{6.3}$$

Proof. The input is the classical Jacobi-sum calculation [3, Theorems 1–2]; we include its character-sum proof. Choose a character χ of order ℓ, and extend all its powers, including the trivial character, by zero at zero. The subgroup indicator gives

$$T_e = \frac{1}{\ell^2} \sum_{i,j=0}^{\ell-1} J(\chi^i, \chi^j), \qquad J(\eta, \theta) = \sum_{x \in \mathbb{F}_q} \eta(x)\theta(1+x).$$

The term $i = j = 0$ is $q-2$. Each of the $2(\ell-1)$ terms with exactly one nontrivial character is -1. The $\ell - 1$ terms with both characters nontrivial and product trivial are also -1: the substitution $u = x/(1+x)$ identifies their sum with the sum of a nontrivial character over $\mathbb{F}_q^\times \setminus \{1\}$.

For completeness, fix a nontrivial additive character ψ and set $G(\eta) = \sum_{x \ne 0} \eta(x)\psi(x)$. For nontrivial η, substituting $x = ty$ into its squared modulus gives

$$|G(\eta)|^2 = \sum_{t \ne 0} \eta(t) \sum_{y \ne 0} \psi((t+1)y) = (q-1) - \sum_{t \ne 0,1} \eta(t) = q.$$

If η, θ, and $\eta\theta$ are all nontrivial, grouping $G(\eta)G(\theta)$ by $z = x + y$ gives

$$G(\eta)G(\theta) = J(\eta, \theta)G(\eta\theta).$$

Here the $z = 0$ contribution vanishes by multiplicative-character orthogonality; for $z \ne 0$ put $x = zu$, $y = z(1+u)$. Thus $|J(\eta, \theta)| = \sqrt{q}$. There are $(\ell-1)(\ell-2)$ remaining terms. Their triangle bound, together with the terms evaluated above, proves (6.1), also when $\ell = 1$.

A pair counted by T_d that does not generate C_d lies in at least one maximal subgroup $C_{d/p}$, for a prime $p \mid d$. Therefore

$$|\mathcal{R}_d| \geq T_d - \sum_{p|d} T_{d/p}.$$

Applying (6.1) to the first term gives $T_d \geq (q+1-3k)/k^2 - b_k\sqrt{q}$. For the subgroup of index kp, the same bound gives

$$T_{d/p} \leq \frac{q+1-3kp}{k^2p^2} + b_{kp}\sqrt{q} \leq \frac{q}{k^2p^2} + \sqrt{q}.$$

This proves (6.2). All the primes dividing d are odd. The elementary telescoping bound

$$\sum_{p|d} \frac{1}{p^2} \leq \sum_{j=1}^{\infty} \frac{1}{(2j+1)^2} < \frac{1}{4} \sum_{j=1}^{\infty} \frac{1}{j(j+1)} = \frac{1}{4}$$

proves (6.3). □

Theorem 6.2 (Realization at any fixed odd index)**.** *Let $k > 1$ be odd. For every sufficiently large n satisfying $k \mid 2^n - 1$, put $q = 2^n$ and $d = (q-1)/k$. Then*

$$H_d = \mathbb{F}_q^+ \rtimes C_d$$

is a primitive subgroup of index k in $\mathrm{AGL}(1,q)$ and is generated by f and $F(f)$ for a scalar affine complete mapping. An explicit sufficient condition is $d > \sqrt{q} - 1$ and

$$L_{q,k,d} := \frac{3q}{4k^2} - \frac{3k-1}{k^2} - (\omega(d) + b_k)\sqrt{q} > 0. \tag{6.4}$$

Under these conditions the number of realizing maps in the specified field coordinates is $(q-1)|\mathcal{R}_d| > (q-1)L_{q,k,d}$. In particular, each fixed odd index $k > 1$ gives an infinite family of realizable proper primitive affine groups.

Proof. The degree $s = \mathrm{ord}_d(2)$ of the field generated by C_d divides n. If $s < n$, then $s \leq n/2$ and $d \mid 2^s - 1 \leq \sqrt{q} - 1$, contrary to the assumed inequality. Hence $s = n$. Proposition 6.1 and (6.4) imply $\mathcal{R}_d \neq \varnothing$. Corollaries 2.5 and 2.4 give the realization, primitivity, and exact count. The index is k because the translation subgroup is full and the complement has order $(q-1)/k$.

To establish the infinite-family assertion without an arithmetic assumption on d, note that $3^{\omega(d)} \leq d < 2^n$. Since $3 > 2^{3/2}$, this yields $\omega(d) < 2n/3$. Also $0 \leq b_k \leq 1$. For fixed k, the positive term $3 \cdot 2^n/(4k^2)$ in (6.4) eventually exceeds the constant term and $(2n/3+1)2^{n/2}$. Likewise $(2^n-1)/k > 2^{n/2} - 1$ eventually. There are infinitely many admissible n, namely multiples of $\mathrm{ord}_k(2)$. The sufficient conditions therefore hold for every sufficiently large admissible dimension. □

6.2 A quantitative divisor sieve and growing indices

Theorem 6.3 (Quantitative slope counts)**.** *Let $q = 2^n$, $d > 1$ divide $q - 1$, and $k = (q-1)/d$. Set*

$$\theta_j(d) = \prod_{p|d} (1 - p^{-j}) \quad (j = 1, 2), \qquad S_{k,d} = \sum_{h|\mathrm{rad}(d)} b_{kh}, \qquad b_\ell = \frac{(\ell-1)(\ell-2)}{\ell^2}.$$

Here $\mathrm{rad}(d)$ is the product of its distinct prime divisors. Then

$$|\mathcal{R}_d| = \frac{q+1}{k^2}\theta_2(d) - \frac{3}{k}\theta_1(d) + E_{q,k,d}, \qquad |E_{q,k,d}| \leq S_{k,d}\sqrt{q} \leq 2^{\omega(d)}\sqrt{q}. \tag{6.5}$$

The sharper error coefficient is

$$S_{k,d} = 2^{\omega(d)} - \frac{3}{k}\prod_{p\mid d}(1+p^{-1}) + \frac{2}{k^2}\prod_{p\mid d}(1+p^{-2}). \tag{6.6}$$

For fixed k, as n tends to infinity through admissible dimensions,

$$|\mathcal{R}_d| = \frac{q}{k^2}\theta_2(d) + O_k(q^{1/2+o(1)}), \qquad \frac{|\mathcal{R}_d|}{T_d} = \theta_2(d) + O_k(q^{-1/2+o(1)}). \tag{6.7}$$

In particular $\theta_2(d) > 3/4$: asymptotically at least three quarters of the slopes satisfying the two subgroup-membership conditions generate the specified complement jointly.

Proof. Write $T_{d/h} = (q+1-3kh)/(kh)^2 + \varepsilon_h$ in (2.1). Proposition 6.1 gives $|\varepsilon_h| \le b_{kh}\sqrt{q}$. Only squarefree h contribute. The identities

$$\sum_{h\mid d}\frac{\mu(h)}{h^j} = \theta_j(d), \qquad \sum_{h\mid \mathrm{rad}(d)} h^{-j} = \prod_{p\mid d}(1+p^{-j})$$

give (6.5) and, since $b_\ell = 1 - 3/\ell + 2/\ell^2$, (6.6). All these coefficients are nonnegative and at most one for integer $\ell \ge 1$.

For each $\epsilon > 0$, choose a fixed cutoff beyond which $2 \le p^\epsilon$. The bounded contribution of smaller primes can be absorbed into a constant, so $2^{\omega(d)} \le C_\epsilon d^\epsilon$. Consequently $2^{\omega(d)} = q^{o(1)}$. The first asymptotic follows from (6.5); the omitted terms are bounded for fixed k. Also $T_d = q/k^2 + O_k(\sqrt{q})$ by (6.1), which gives the ratio assertion. Finally $\prod(1-x_i) \ge 1 - \sum x_i$ for $0 \le x_i \le 1$, and the odd-prime bound in Proposition 6.1 is strictly less than $1/4$. Thus $\theta_2(d) > 3/4$. □

Corollary 6.4 (A uniform range of growing indices)**.** *Let $n \ge 16$, $q = 2^n$, and $k > 1$ divide $q-1$. If*

$$k \le \frac{q^{1/4}}{\sqrt{n}}, \tag{6.8}$$

then $H_{(q-1)/k}$ is a realizable proper primitive affine companion group.

Proof. Put $d = (q-1)/k$. Since $k < \sqrt{q}+1$, we have $d > \sqrt{q}-1$, so the field-degree argument in Theorem 6.2 gives $\mathrm{ord}_d(2) = n$. The index is odd and at least three; hence $(3k-1)/k^2 < 1$. Using $\omega(d) < 2n/3$ and $b_k \le 1$ gives

$$L_{q,k,d} > \left(\frac{n}{12} - 1\right)\sqrt{q} - 1 > 0 \qquad (n \ge 16).$$

Theorem 6.2 proves the assertion. □

6.3 Subfield descent and the quarter-power barrier

Theorem 6.5 (An infinite family of primitive nonrealizable groups)**.** *Let $r \ge 2$, $q = 2^{2r}$, and $d = 2^r + 1$. The scalar affine group*

$$H_d = \mathbb{F}_q^+ \rtimes C_d \le \mathrm{AGL}(1,q)$$

is primitive of degree q, but no affine complete mapping f on $\mathbb{F}_2^{2r}$ satisfies $\langle f, F(f)\rangle = H_d$. The same nonrealizability holds for every $\mathrm{GL}(2r,2)$-conjugate of this subgroup. More generally, if an odd integer $d > 3$ divides $2^r + 1$, then $\mathcal{R}_d = \varnothing$ in every characteristic-two field containing C_d.

Proof. Since $2^r \equiv -1 \pmod{d}$, the order of 2 modulo d divides $2r$. It cannot be at most r, as $0 < 2^j - 1 < d$ for $1 \le j \le r$. Every proper divisor of $2r$ is at most r, so $\mathrm{ord}_d(2) = 2r$. Consequently the multiplicative action of C_d on $\mathbb{F}_q$ is irreducible over $\mathbb{F}_2$, and its affine extension by all translations is primitive.

More generally, suppose $d > 3$ divides $2^r + 1$ and $a, 1 + a \in C_d$. Because $a^d = 1$ and $(1 + a)^d = 1$, we have $a^{2^r} = a^{-1}$ and

$$1 = (1 + a)^{2^r+1} = (1 + a^{-1})(1 + a) = a + a^{-1}.$$

Multiplying by a yields $a^2 + a + 1 = 0$. Thus a and $1 + a = a^2$ both have order 3, and $\langle a, 1 + a\rangle = C_3$. Since $d > 3$, no a belongs to $\mathcal{R}_d$. Corollary 2.5 proves that no scalar affine companion pair generates H_d. Moreover, every affine permutation belonging to the specified group H_d has precisely the scalar form $x \mapsto ax + c$, so this excludes arbitrary affine complete mappings whose generated group equals H_d.

Finally, for an invertible $\mathbb{F}_2$-linear change of coordinates S, one has $F(SfS^{-1}) = SF(f)S^{-1}$. Hence an affine companion realization of a conjugate subgroup would conjugate back to one of H_d, which is impossible. □

The theorem gives an obstruction beyond irreducibility and primitivity; it does not claim that all nonrealizable primitive affine groups have a complement of this form.

Theorem 6.6 (Norm descent into a proper subfield)**.** *Let $1 \le s < r$, $s \mid r$, $u \mid 2^s - 1$, and work in $L = \mathbb{F}_{2^{2r}}$. Put*

$$d = u(2^r + 1), \qquad g = \begin{cases} u, & r/s \text{ even}, \\ u(2^s + 1), & r/s \text{ odd}. \end{cases}$$

The two membership conditions descend exactly:

$$\{a \in C_d : 1 + a \in C_d\} = \{a \in C_g \subset \mathbb{F}_{2^{2s}}^{\times} : 1 + a \in C_g\}. \tag{6.9}$$

Thus $T_d = T_g$, $\mathcal{R}_d = \varnothing$, and $H_d = L^+ \rtimes C_d$ is primitive but cannot be generated by an affine complete mapping and its prescribed companion. The nonrealizability also holds for every linear conjugate of H_d.

Proof. Set $Q = 2^r$, $K = \mathbb{F}_Q$, and $E = \mathbb{F}_{2^s} \subset K$. The norm $N_{L/K}(x) = x^{Q+1}$ has kernel C_{Q+1} and maps C_d onto $C_u \subset E^{\times}$: its image has order $d/\gcd(d, Q + 1) = u$, since $\gcd(u, Q + 1) = 1$. If $a, 1 + a \in C_d$, put $U = N(a)$ and $W = N(1 + a)$. Both lie in E, and

$$a + a^Q = 1 + U + W \in E, \qquad a^2 + (1 + U + W)a + U = 0.$$

Thus a has degree at most two over E and belongs to $\mathbb{F}_{2^{2s}} \subset L$. The same holds for $1 + a$.

Write $t = 2^s$ and $m = r/s$. Cyclic subgroup intersection gives $C_d \cap \mathbb{F}_{t^2}^{\times} = C_{\gcd(d,t^2-1)}$. If m is even, $t^m + 1 \equiv 2 \pmod{t^2 - 1}$, so this gcd is u. If m is odd, $t^m + 1 \equiv t + 1 \pmod{t^2 - 1}$; since $u \mid t - 1$ and $\gcd(t - 1, t + 1) = 1$, the gcd is $u(t + 1)$. This proves one inclusion in (6.9); the other follows from $C_g \subset C_d$. In both cases $g < d$, so every pair in the left-hand set generates a proper subgroup of C_d. Therefore $\mathcal{R}_d = \varnothing$.

The group C_d contains C_{2^r+1}, whose action on L is irreducible by the field-degree proof in Theorem 6.5. Hence H_d is primitive. Corollary 2.5 excludes scalar realizations, and every affine permutation in the specified group is scalar. Linear-conjugacy equivariance of F excludes realizations of its linear conjugates as well. □

Corollary 6.7 (The quarter-power barrier)**.** *For every $s \geq 1$, put $t = 2^s$, $q = t^4$, and*

$$k = t + 1 = q^{1/4} + 1, \qquad d = (t-1)(t^2+1) = \frac{q-1}{k}.$$

Then H_d is a proper primitive nonrealizable affine companion group, and its membership pairs are exactly

$$\{a \in C_d : 1 + a \in C_d\} = \mathbb{F}_t \setminus \{0, 1\}. \qquad T_d = t - 2, \quad |\mathcal{R}_d| = 0. \tag{6.10}$$

Consequently no exponent $\beta > 1/4$ can give universal eventual realizability for all admissible indices $1 < k \leq q^\beta$. Even the range $1 < k \leq Cq^{1/4}$ fails for every fixed $C > 1$.

Proof. Apply Theorem 6.6 with $r = 2s$ and $u = t - 1$. Here $g = t - 1$ and $C_g = \mathbb{F}_t^\times$, yielding (6.10). The indices are proper and $k/q^{1/4} = 1 + 1/t \to 1$. Hence these infinitely many exceptions eventually lie in either range in the statement. □

Together, Corollaries 6.4 and 6.7 determine the exponent $1/4$ for a universal growing-index guarantee. They leave a factor of $\sqrt{n} = \sqrt{\log_2 q}$ between the sufficient range and the obstructed family. In particular, the exact boundary $k \leq q^{1/4}$ has not been decided by these arguments.

7 Scope and further questions

The norm-fiber method separates three scales: the $O(t)$ population of one fiber, the $O(t^2)$ population of the full intersection, and the $O(t)$ trace-one subfamily used for all-parameter existence. The populations below make this distinction concrete. Here T_d counts subgroup membership, N_d^{prim} requires only a to have order d, and Δ_t is the Frobenius-coupled subfamily.

s	T_d	$\lvert\mathcal{R}_d\rvert$	N_d^{prim}	D_d	Δ_t
2	24	24	16	8	8
3	86	72	48	24	0
4	240	240	224	208	16
5	1214	960	520	240	0
6	4104	3960	2928	1992	24

These are exact finite counts. The asymptotic and all-parameter claims follow from the proofs, not from this table. The single full-order estimate, the four-torsion symmetry of good fibers and its separate existence proof are retained in Supplementary Section S6.

The relative-trace refinement gives more than this finite table: Theorem 5.1 controls all M_c, and Theorem 5.4 classifies every empty nonzero-relative-trace slice in E^*. The exceptional traces form C_9 at $s = 3$ and C_{11} at $s = 5$. At $s = 3$, reciprocal traces of orders $1, 3, 9$ cannot support the required order 65. At $s = 5$, the quotient by C_{11} forces reciprocal-trace order 3 or 33, and the surviving order-33 parameters lose a factor 5 in their order. The two earlier trace-one exceptions are the intersections of these groups with K^*. The inverse-support reduction makes a common organization explicit: a missing primitive support gives the whole empty C_9, whereas a nonempty support confined to one prime-coordinate kernel gives the empty C_{11}. For the other small norm classes, representative positivity and one separation witness exclude the only possible two-prime obstruction. This is a proof reduction within a classical cyclic-group framework; it does not supply a new general character estimate or an unrestricted higher-prime classification.

The refinement also gives information beyond existence. Proposition 5.5 determines the mean on every fixed-norm class of traces and bounds the variance without the $W(A)$ factor in the pointwise

sieve. At $t = 32$, the support $T_1 = C_{11} \setminus \{1\}$ explains both the empty subgroup and the two positive populations, 20 and 18. The zero set is a subgroup, but the count is not constant on its other cosets. For arbitrary nonzero K-norm, the exceptional sets become the scalar cosets in Corollary 5.7. This is a transfer of the norm-one result, not an independent existence method.

Three questions remain open in the present treatment. First, the sufficient uniform range and the obstructed family leave a factor $\sqrt{\log_2 q}$; the exact universal boundary for admissible $k \le q^{1/4}$ is unresolved here. Second, positivity in every primitive norm fiber is proved in the explicit asymptotic range, whereas full-family positivity at small parameters is weaker. A sharp description of exceptional small norm fibers would refine the distribution theorem. This is different from relative-trace slices, whose complete exceptional classification is now given above. Third, the argument uses the genus-one geometry of a quartic tower. Extending the norm-dependent coefficient calculation to higher towers requires a corresponding analysis of singular fibers and the geometric character exceptions.

The companion-group reduction applies to all affine complete mappings. Its linear complement is abelian, so the resulting groups are metabelian. The supplement records the genuinely different nonlinear behavior: an explicit companion pair generates A_{16} (Proposition S4.2). Its coordinate examples and oriented-spread classifications provide further applications of F, with separate statements of computational coverage.

A Finite certificates and the all-parameter proof

Theorem 5.4 reduces the all-parameter conclusion for D_d to the field certificates at $s = 3, 5$. The tables below give the larger verified range $2 \le s \le 32$ as additional exact checks. The integer bounds later in this appendix also establish uniform nonzero-trace positivity for $s \ge 33$. The first table certifies one full-order slope and the additional exponent table certifies both linked slopes; both tables are retained to make their distinct roles explicit. A hexadecimal integer encodes a binary polynomial with its least significant bit the constant coefficient. Each row means

$$E = \mathbb{F}_2[X]/(M_s), \qquad L = E[Z]/(Z^2 + (1 + b + w)Z + b), \qquad a = Z.$$

Thus only M_s, b, w need be specified; all field operations are exact.

s	M_s (hex)	b (hex)	w (hex)
2	13	8	1
3	43	6	3b
4	11b	2f	1
5	409	9b	2c9
6	1009	c78	8e5
7	4021	3e8a	1
8	1002b	2878	689
9	40009	303be	12d39
10	100009	a3085	1
11	400003	662cc	3205f9
12	100001b	d0842d	1
13	400001b	32f9317	1
14	10000003	15a3baa	6c6a1f9
15	40000003	317d6a85	36559252
16	10000008d	4893e13d	2e37b719
17	40000001b	dcbd299	91607c0b

s	M_s (hex)	b (hex)	w (hex)
18	1000000035	78c0283c4	1
19	4000000063	1153a3d1ef	16004f10d3
20	10000000039	1f9215dca7	1
21	40000000027	2657c0cd035	374caba3695
22	100000000021	fe622270c1c	1
23	400000000003	17c45ffbd76	29bd79b1ae4
24	100000000002d	fed891dbe6c9	ec7a6a1920af
25	400000000001d	2907b6c341c09	1
26	10000000000009	c6d27ccd2b0df	432bf4829f702
27	4000000000007d	28047cef28c34e	1
28	100000000000095	a586528c5b1c9	1
29	400000000000063	243f16ec4e21119	1
30	1000000000000003	8c388b10c413f59	1
31	4000000000000069	12dfcd22c3b1728b	1
32	1000000000000001b	ae3111b3bb81428f	c171e75cc47c2b2e

The ancillary file `inner_critical_certificates.json` also records the full factorizations of A, B, every nonidentity residue $a^{d/p}$ for a prime $p \mid d$, and recursive Lucas primality certificates for those primes. Verification consists of the following finite identities, carried out by `verify_inner_critical.py`. The additional `verify_inner_symmetry.py` checks the four-torsion identities and rational Lang-fiber multiplier at all 31 certificate points, and enumerates every primitive-norm fiber for $s = 2, \ldots, 6$. It checks four-point orbits, order preservation, divisibility by four and the exact integer multiplicity distributions underlying the character symmetries. The new verification logs and runtime metadata are supplied with the ancillary files; these finite checks supplement the paper proof of Proposition S6.2. First, M_s has degree $n = 2s$, and

$$X^{2^n} \equiv X \pmod{M_s}, \qquad \gcd(M_s, X^{2^{n/r}} - X) = 1 \quad (r \mid n \text{ prime}),$$

which certify irreducibility. Second, $b^A = w^A = 1$, $b^{A/p} \neq 1$ for every prime $p \mid A$, $c = 1 + b + w \neq 0$, and $\mathrm{Tr}_{E/\mathbb{F}_2}(b/c^2) = 1$. Third, in the quadratic field one checks

$$a^d = (1+a)^d = 1, \quad a^B = b, \quad (1+a)^B = w, \quad a^{d/p} \neq 1 \quad (p \mid d \text{ prime}).$$

Primality is certified without probable-prime assumptions: the base case is 2; for an integer $p > 2$ one supplies a complete recursively certified factorization of $p - 1$ and an integer v with $v^{p-1} \equiv 1 \pmod{p}$ and $\gcd(v^{(p-1)/r} - 1, p) = 1$ for every prime $r \mid p - 1$. For any prime divisor ℓ of p, these conditions force the order of v modulo ℓ to be $p - 1$, hence $\ell \geq p$ and p is prime. The independent verifier uses different field multiplication and powering implementations from the certificate finder; it requires only the Python standard library, run without the `-O` flag.

The small-prime list used in the analytic constant is

$$5, 13, 17, 29, 37, 41, 53, 61, 73, 89, 97, 101,$$
$$109, 113, 137, 149, 157, 173, 181, 193, 197, 229, 233, 241.$$

Their product is 6045408114863853991373345088496297320256672385; the integer comparison is $256^{24} < 32^8$ times this product. The verifier checks this comparison as well.

As a separate small-field audit, multiplicative-subgroup enumeration and polynomial gcds with

the Möbius sieve agree on the following counts. Here N_d^{prim} has the meaning in Theorem S6.3.

s	d	T_d	$\lvert\mathcal{R}_d\rvert$	N_d^{prim}	D_d
2	85	24	24	16	8
3	585	86	72	48	24
4	4369	240	240	224	208
5	33825	1214	960	520	240
6	266305	4104	3960	2928	1992

The data also verify the genus-one Hasse bounds on every primitive norm fiber in this small range. They confirm that the diagonal $b = w$ gives no realizing slope for $s = 3, 5$. These supplementary counts do not replace the general proof. The present all-dimensional theorem uses the classical Weil bound and a finite exact certificate, and is outside the Lean formalization.

Additional double-order certificates. The file `inner_double_order_certificates.json` uses the same M_s, b as the preceding table and replaces w by b^{e_s}, with the following exponents. The root $a = Z$ of $Z^2 + (1 + b + b^{e_s})Z + b$ and its translate $1 + a$ both have exact order d, for every $2 \le s \le 32$.

s	e_s	s	e_s
2	1	18	1
3	4	19	5
4	1	20	1
5	4	21	5
6	3	22	1
7	1	23	8
8	4	24	4
9	26	25	1
10	1	26	3
11	7	27	1
12	1	28	1
13	1	29	1
14	8	30	1
15	4	31	1
16	4	32	3
17	7		

The independent `verify_inner_mixed.py` checks irreducibility, the trace, both norms and both nonidentity residues $a^{d/p}, (1 + a)^{d/p}$ for every prime $p \mid d$. It uses the complete factorizations and recursive Lucas certificates of `inner_critical_certificates.json`; all new residues and all norm-fiber counts for $s = 2, \ldots, 6$ are recorded in `inner_mixed_verification.json`. It also checks the corrected mixed Lang functions and trace multipliers at the 31 certificate points. These computations establish precisely the stated finite range. Together with Proposition S8.1 for $s \ge 33$, they complete the all-dimensional double full-order theorem. The new verifier `verify_inner_double_all.py` independently rechecks all 31 double-order certificates, including their recursive Lucas primality proofs, without calling a certificate finder.

Integer bounds for double full-order existence

The bound in Proposition S8.1 is positive when $\sqrt{t} \ge 4W(d)$. We cover $33 \le s \le 113$ by exact divisor upper bounds, and $s \ge 114$ by a uniform inequality for odd integers. No large residual factor is assumed to be prime.

For the uniform bound let P be the product of all 53 odd primes less than 256, namely

$$3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83,$$
$$89,97,101,103,107,109,113,127,131,137,139,149,151,157,163,167,173,$$
$$179,181,191,193,197,199,211,223,227,229,233,239,241,251.$$

The exact integer comparison $256^{53} < 4096^8 P$ gives, for every odd n,

$$\frac{W(n)^8}{n} = \prod_{p^e \| n} \frac{256}{p^e} \le \prod_{3 \le p < 256} \frac{256}{p} = \frac{256^{53}}{P} < 4096^8.$$

Here the product on the right is over primes: each factor for a prime at least 256 is at most one, and adding missing smaller prime factors only increases the bound. Thus $W(n) < 4096 n^{1/8}$. For $s \ge 114$, $d < 2t^3$ and $t > 2^{113}$ prove the sufficient condition, as computed in Proposition S8.1.

For each $33 \le s \le 113$, divide A and B by every prime at most 4096, with its full multiplicity. Let j_A, j_B be the numbers of distinct small factors and R_A, R_B the residuals. All prime factors of each residual exceed 4096. If $4097^{k_A} \le R_A < 4097^{k_A+1}$ and similarly for B, then

$$\omega(d) = \omega(A) + \omega(B) \le r_s := j_A + k_A + j_B + k_B.$$

The following table lists all 81 bounds; each satisfies $s \ge 4 + 2r_s$, so $\sqrt{t} \ge 4\,2^{r_s} \ge 4W(d)$.

s	r_s	s	r_s	s	r_s
33	11	60	15	87	24
34	9	61	16	88	21
35	13	62	15	89	23
36	11	63	21	90	27
37	11	64	15	91	26
38	11	65	20	92	23
39	13	66	19	93	25
40	9	67	18	94	24
41	11	68	16	95	26
42	14	69	19	96	24
43	12	70	20	97	26
44	11	71	18	98	26
45	17	72	19	99	29
46	13	73	20	100	26
47	13	74	19	101	26
48	12	75	23	102	27
49	15	76	19	103	26
50	14	77	23	104	25
51	15	78	22	105	32
52	12	79	21	106	27
53	14	80	19	107	28
54	16	81	23	108	29
55	16	82	21	109	27
56	13	83	22	110	29
57	16	84	23	111	30
58	15	85	24	112	27
59	15	86	22	113	29

The full small factorizations, residual integers and their integer power bounds are supplied in `inner_double_omega_certificates.json`. The independent verifier `verify_inner_double_all.`

py reconstructs every small prime by deterministic trial division, checks the exact products and all 81 inequalities, and checks $256^{53} < 4096^8 P$. The source results and environment record are `inner_double_all_verification.json` and `double_all_verification_environment.json`. The reorganization includes a fresh finite-verification log at `verification/author_runtime/author_double_order_audit.log`. It also exhaustively checks the trace-one subfamily for $2 \le s \le 6$: its counts are respectively $8, 0, 16, 0, 24$. In particular the all-dimensional claim is for D_d, not for Δ_t. These older field certificates and the integer bounds remain valid. Appendix B gives the stronger prime-sieve range and the complete trace-one classification.

B Certificates for all relative traces and the exceptional groups

For Lemma 5.3, the following table lists a squarefree core e and an integer upper bound $C_s \ge 4W(e)\Lambda - 2$. In every row $\delta > 0$ and $(2^s+1)^2 > 2^s C_s^2$, so positivity holds for every $c \in E^*$. The omitted primes are all prime divisors of $(2^s+1)(2^{2s}+1)$ outside e. Full factorizations, the exact rational δ, Λ and both integers in each comparison are recorded in `verification/uniform_trace/uniform_trace_checks.json`.

s	e	C_s	s	e	C_s
8	1	11	21	3	100
10	1	23	22	1	34
11	1	41	23	1	59
12	1	24	24	1	27
13	1	52	25	3	98
14	1	29	26	1	40
15	3	96	27	3	103
16	1	15	28	1	24
17	1	50	29	1	51
18	1	52	30	1	58
19	1	50	31	1	58
20	1	19	32	1	19

No asymptotic extrapolation from this table is used: $s \ge 33$ is supplied by the separate integer bounds of Appendix A and Theorem 5.1.

For the finite trace-one counts, define H_A as in (5.12). In the next table an entry $e : j$ means $\deg \gcd(H_A, X^{B/e}+1) = j$; only nonzero degrees are listed. Every unlisted squarefree divisor of B has degree zero. Möbius inversion gives the last column.

s	$\deg H_A$	Nonzero gcd degrees $e : j$	Δ_t
2	8	$1:8$	8
3	12	none	0
4	32	$1:16$	16
5	40	$1:20,\ 5:20$	0
6	96	$1:24$	24
7	168	$1:84$	84
8	512	$1:288$	288
9	648	$1:360,\ 5:36,\ 13:72,\ 37:36$	216

For the two exceptions the reciprocal-trace polynomials are encoded by the hexadecimal integers

$$H_9 = \texttt{13f9}, \qquad H_{33} = \texttt{1950fbbe153}.$$

The verifier constructs the cyclotomic polynomials by exact division, computes every modular power and gcd, and checks explicit polynomial identities $uH_A + v\Phi_B = 1$ at $s = 3, 5$. The coefficients u, v are archived as binary-polynomial integers. It additionally checks the degree-20 order drop at $s = 5$ and the coprimalities excluding orders 5 and 41.

The standard-library verifier is `verification/uniform_trace/verify_uniform_traces.py`. Its independent small-field checks enumerate all nonzero K-valued relative traces for $2 \le s \le 6$ in two ways: through the full norm-one subgroup and through the reciprocal-trace map on V. They agree on both Q_c and M_c, on the binary Kloosterman identity and on all stated numerical error bounds. The analytic character estimate and the infinite parameter range rely on the proofs, not on these finite checks.

Representative and separating support witnesses

Write $\rho = \mathrm{rad}(A)$ and choose generators $z \in E^*$, $\Gamma \in V$. Represent $h = z^{Aj}$, $0 \le j < t - 1$. Each certificate records k and e with

$$\gcd(k, B) = 1, \qquad \Gamma^k + \Gamma^{-k} = h^2(z^{t-1})^e.$$

A central witness has $\gcd(e, A) = 1$ and proves $M_h > 0$. When A has two distinct prime divisors, a separating witness has exponent e' with $e' \not\equiv e, -e \pmod{\rho}$. Inversion supplies the second residue $-e$, so these two witnesses certify at least three distinct points in the radical quotient. The inverse-support lemma then proves positivity on the whole trace class hU. Only these witnesses, not the complete support populations, are required by this step of the proof.

s	A	ρ	Central witnesses	Separating witnesses
2	5	5	3	0
3	9	3	6	0
4	17	17	15	0
5	33	33	30	30
6	65	65	63	63
7	129	129	127	127
9	513	57	511	511
Total			755	731

The omitted representatives are precisely $(s, h) = (3, 1), (5, 1)$, whose supports are determined by the polynomial identities. The case $s = 8$ is already analytic; 255 additional central witnesses are retained as a finite check. Thus the archive contains $1010 + 731 = 1741$ witnesses, of which 1486 are used by the finite part of the current proof.

The exact data are in `verification/radical_support_2026-10-26/support_certificates.json`. The producer enumerates reciprocal traces from primitive γ and records central and separating exponents. The independent verifier uses single binary polynomial fields of degree $4s$, checks irreducibility and the field embeddings, and directly verifies every reciprocal trace and order condition. It constructs a full-order element for each central witness. The two exceptional supports are separately enumerated at $s = 3, 5$. Neither verification nor the group-theoretic proof uses the truth of the completed trace classification.

The older all-E certificate family below is retained for cross-checking. It already compressed traces into witness families; the improvement here is the reduction to representative and separating support witnesses, without checking every trace residue for coverage. Excluding the analytic $s = 8$

case, the older proof used 2809 family records, compared with 1486 support witnesses in the present finite argument.

Retained witness families on the whole trace field

Put $r = t - 1$ and $N = t^2 - 1 = rA$, and choose a primitive $z \in E^*$ and a generator $\Gamma \in V$. Write every $c \in E^*$ uniquely as $c = z^{\ell + rm}$, with $0 \le \ell < r$ and $0 \le m < A$. For a recorded exponent k with $\gcd(k, B) = 1$, certify

$$\Gamma^k + \Gamma^{-k} = z^j, \qquad j \equiv 2\ell \pmod{r}.$$

Then $w = z^{j - 2\ell - 2rm} \in U$, and it has order A exactly when

$$\gcd\left(\frac{j - 2\ell}{r} - 2m, A\right) = 1.$$

For each such residue m, Theorem 5.1 constructs a full-order element $a = c/(1 + \Gamma^k)$. The recorded lists of k cover every residue outside the two exceptional groups, as follows. "Families" counts recorded values of k; it is not a count of individual trace slices.

s	$\lvert E^*\rvert$	Nonempty traces	Families	Empty traces
2	15	15	6	none
3	63	54	12	C_9
4	255	255	30	none
5	1023	1012	117	C_{11}
6	4095	4095	227	none
7	16383	16383	484	none
8	65535	65535	510	none
9	262143	262143	1933	none

The coefficients, primitive generators, logarithms j and lists are in `verification/extended_trace_2026-10-26/extended_trace_certificates.json`. The generator uses the exact recurrence $\tau_{k+1} = \tau_1 \tau_k + \tau_{k-1}$ for $\tau_k = \Gamma^k + \Gamma^{-k}$. The independent verifier instead works in single binary polynomial fields of degree $4s$. It checks their irreducibility and the field isomorphisms, verifies every displayed logarithm by direct powering, and tests every coverage residue by the gcd condition above. For each of the 3319 families it also constructs an element and directly verifies its relative trace, norm and exact order. Thus 349492 nonempty traces are certified; the remaining $9 + 11$ traces are proved empty in Theorem 5.4.

For $s = 2, \dots, 6$, independent enumeration of the entire subgroup checks all 5451 nonzero E-valued populations, both stated error bounds and $Q_c = Q_h$ for $h^2 = N_{E/K}(c)$. It also confirms that M_c need not equal M_h. The verifier and exact output are `verification/extended_trace_2026-10-26/verify_extended.py` and `independent_checks.json` in that directory.

The earlier 1010 individual witnesses for the K^* restriction remain archived in `verification/uniform_trace/`, with their separate binary-field audit in `verification/proof_audit_51_54_2026-10-26/`. The trace-one polynomial counts and integer tail certificates are retained. Positivity in the infinite range follows from the analytic proof, not from extrapolating these finite checks.

C Data, verification and formalization scope

The accompanying project contains the exact field certificates, deterministic verifiers and current sources. The archived input and its original verification materials come from

https://github.com/63331849121/yijing-mathematical-structures.

The archived input is the directory PaperA_scalar_realizations at revision d33c51696f81492de67b8df2166d0c887666176c. The archive records file hashes and the correspondence between the original and present theorem labels.

The verifier verify_inner_double_all.py checks the finite double-order certificates and the integer inequalities used in the all-parameter proof. Appendix A gives the complete parameter partition. A separate exact-arithmetic audit checks all 31 field certificates for $2 \le s \le 32$, all 81 intermediate divisor bounds for $33 \le s \le 113$, and the full small populations in Section 7. The project verification files state the inputs and outcomes. These finite computations support the stated certificates; the infinite ranges depend on the mathematical arguments.

The earlier K-valued relative-trace verifier and its exact output are in verification/uniform_trace/. They check all 24 prime-sieve rows, all eight trace-one polynomial counts, the two Bézout identities for emptiness, and every nonzero K-valued trace slice at $s = 2, \dots, 6$ by independent parameterizations. They also verify 1010 full-order witnesses covering every nonexceptional nonzero K-valued slice at $s = 2, \dots, 9$. The current finite proof for the whole E^* uses the files in verification/radical_support_2026-10-26/. It has 755 central witnesses and 731 separating witnesses; 255 additional central witnesses at the analytic case $s = 8$ are redundant checks. A separate arithmetic implementation transports the data to degree-$4s$ binary polynomial fields and verifies every witness directly. The inverse-support lemma propagates positivity from the representatives to the whole trace classes. The two exceptional supports are independently checked as well. The general reduction depends only on Theorem 5.1 and elementary cyclic-group arguments, not on the completed classification. Exhaustive small cyclic-group checks and counterexamples to dropping its hypotheses are also recorded.

The earlier 3319 exact witness families in verification/extended_trace_2026-10-26/ remain a separate check on all 349512 nonzero traces for $s = 2, \dots, 9$. Independent full-subgroup enumeration for $s = 2, \dots, 6$ additionally checks all 5451 nonzero E-valued populations and the stated error bounds. The phase reduction and prime sieve prove the infinite range; finite checks do not replace those arguments. Appendix B distinguishes the current support proof from the retained all-trace certificates.

The independent script in verification/fulltext_structure_2026-10-26/verify_structure.py enumerates all nonzero ambient-field elements for $s = 2, \dots, 5$. It checks the convolution on all 1356 nonzero relative traces, all 36080 norm–trace scaling cases (including trace zero), the exact means, the variance inequalities and the radical periods. Separate binary-polynomial divisions verify the exceptional reciprocal-trace supports. These are independent finite checks; the general Fourier and scaling identities are proved in Section 5.

The rational-image comparison with [27] has a further independent verifier in verification/cstc26_fulltext_2026-10-26/verify_specialization.py. It checks the exact inverse-trace summation on all 5451 nonzero relative traces for $s = 2, \dots, 6$, including the 20 empty slices, and evaluates the rational function on every full-order element in these norm kernels. It also checks the specialized sufficient inequalities by exact rational arithmetic.

The comparisons with [10, 11] have exact checks in verification/older_sources_2026-10-26/verify_older_sources.py. For $s = 2, \dots, 9$ they evaluate every squarefree-core partition for the direct prescribed-order count and its primitive lift. The all-parameter obstruction follows from the inequalities in Appendix D. The script also matches the four published binary rows $t = 4, 8, 16, 32$ of [11, Table 1] against the preceding independent norm-kernel enumeration, and separately checks the distinct binary rational-function example in [10]. These are finite consistency checks, not a new full enumeration of the quartic fields or a proof of global priority.

The archived Lean files cover function-level triality, unique fixed points, the field-Singer generation argument and a conditional Frobenius/order interface. They do not formalize the general matrix-group formula, norm-fiber character estimates, density theorems or all-parameter existence. Supplementary Appendix S.B records their pinned Lean and mathlib versions and the original verification scope. The current revision includes no complete Lean formalization of the paper. The three original Lean files were separately rerun in an isolated environment with the pinned toolchain and dependency commits; all passed, using the existing mathlib archive cache. Two hashes in the historical check record differ from the distributed sources, so the new actual hashes and successful runs are recorded separately in `verification/supplement_sync_2026-10-26/`.

The new dependency-free module `formalization/InverseSupportGrid.lean` freshly checks six lemmas forming the two-coordinate combinatorial core of the inverse-support reduction. Their axiom audit is empty. The cyclic-group CRT and finite-field interfaces, field certificates and geometric estimates remain paper proofs or external exact-arithmetic checks. In particular this module does not formalize the complete trace classification. Supplementary Appendix S.B gives the precise current and historical formalization scopes.

D Comparison with earlier prescribed-order results

We compare the relative-trace theorems with the closest prescribed-order, trace and primitive-normal frameworks. Throughout, $R = t-1$, $A = t+1$, $B = t^2+1$ and $d = AB$. The issue is the effective error and the complete classification of individual traces, since the counting object itself is prior.

D.1 Prescribed order and trace

The counting problem itself has direct predecessors. With $q_0 = t^2$, $n = 2$, $\ell = t-1$, one function $f(X) = X$ and prescribed trace c, it is exactly $N_{\ell,1}$ of [22, §2]; it is also $P_{t^2,2,d}(c)$ of [23, §6]. The zero additive-frequency term in [22, eq. (2.2)] is

$$\frac{\varphi(d)}{t^2} = \theta(d)(t+1) + \theta(d)(t^{-1}+t^{-2}),$$

so the leading linear main term is already present in that framework. However, the bracket in its Theorem 2.4 specializes to $1-2(t-1)W(d) < 0$. That bound gives no positivity here; (5.5) obtains the smaller error scale $W(d)\sqrt{t}$ by using the curve over K.

One can also lift $a = \beta^{t-1}$ with β primitive in L, apply the primitive sieve to $f(X) = X^{t-1}$, and divide the count by $\varphi(t-1)$. In the notation of [22, Theorem 3.1], this gives $D = (q_0-1)(t-1)$, $E = q_0 - 1$ and $\delta_n = 0$. Its sufficient inequality would imply

$$q_0 > (q_0-1)(t-2) + q_0^{-1} > q_0 \qquad (t \geq 4),$$

since its squarefree-core sieve factor is at least one. Thus neither of these direct specializations supplies the present conclusion. This comparison concerns the stated bounds, rather than every refinement of their methods.

The earlier function-field framework of [10] includes the same count. In its Theorem 4.3 take $q = Q = t^2$, $k = 1$, $m = 2$, $r = 1$, $f_1(X) = X$ and index $a = R = t-1$. Strong linear independence holds because each nonzero linear combination has a simple pole. Write u, v for its two coprime squarefree sieve factors, so $uv = \mathrm{rad}(d)$, and put $z = uR \geq 3$. Each linear Artin–Schreier subcover has genus zero. Its displayed sufficient condition therefore requires

$$\theta(v) > 1 - \frac{1}{Q} + \frac{2(Q-1)(z-1)Q + Q(2+z) - 2}{Q(Q^2-1)} > 1,$$

which is impossible for every allowed core.

The rough genus bound is not essential to this failure. Following [10, Remark 4.4 and proof of Theorem 4.3], retain the exact genus of $y^Q - y = \alpha^{vR} x^{uR} + \delta$. Since z is odd, it is $(Q-1)(z-1)/2$. The improved threshold becomes

$$1 - \frac{1}{Q} + \frac{z-1}{Q+1} + \frac{Q(2+z)-2}{Q(Q^2-1)} > 1.$$

Indeed $z \geq 3$ and $2/(Q+1) > 1/Q$. This also rules out the primitive lift: use index 1 and $f_1(X) = X^R$, and divide its count by $\varphi(R)$. Its exact composed-cover genus is the same. Encoding the E-valued trace by two K-valued traces also gives no improvement: take $q = t$, $k = m = 2$, $r = 2$ and functions $X, \xi X$, with $\xi \in E \setminus K$. The trace pairing makes these two coordinates equivalent to the prescribed relative trace. The cover still has degree $q^r = Q$ and the same exact genus, while the pole-degree correction increases. Thus neither the choice of sieve core nor this change of trace coordinates repairs the criterion.

The asymptotic statements [10, Theorems 4.5–4.6] do not supply the missing uniform conclusion. Theorem 4.6 does allow a growing order index, but fixes the coefficient field and the trace functions, including their number. Its large-extension assertion does not specify a relative trace in a growing quadratic subfield. Re-encoding that value by binary traces requires a growing number of functions, outside those fixed-data quantifiers.

There is also a superficial coincidence of exceptional exponents. In [10, Example 5.1 and Theorem 5.2], the primitive element lies in $\mathbb{F}_{2^m}$ and satisfies two absolute-trace conditions for fixed rational functions; the exceptional exponents include $3, 5$. Taking $m = s$ gives order $t-1$ in K, whereas the present problem has order d in L. Taking $m = 4s$ gives the ambient field L but a different order and different trace constraints. At $s = 2$ that example has no admissible primitive element, whereas $M_1 = 8$ here. The two existence statements are therefore different; no conversion to the relative-trace classification is supplied by those results.

The line framework of [6, criterion (9)] also counts M_c: use base E, extension degree two, index $t-1$, and the affine line of relative trace c. Its displayed sufficient condition becomes $t > (t-1)W(d)$, which fails for $t \geq 4$. Thus the counting object and the line interpretation are prior.

D.2 Product, hypersurface and rational-image criteria

The product criterion in [25, Theorem 1.5(1)] fails for every allowed decomposition of the index, not just a selected one. Use base E and extension degree two. Put $R = t-1$, $m = t^4 - 1$, and $\vartheta = \varphi(m)/m$. For $R = r_1 + \cdots + r_k$ with all $r_i \mid m$, their notation gives $\Lambda_{r_i} = \varphi(m)/\varphi(m/r_i)$ and $\Lambda_R = \varphi(R)$. Every r_i is odd, so $k \geq 3$ is odd. Each $a_i = \vartheta/\Lambda_{r_i}$ satisfies $a_i \leq \vartheta \leq 2/3$ and $a_i \leq 1/r_i$. One r_i is at least R/k, and hence the squared ratio of the two sides of their required inequality is at most

$$\frac{\vartheta^{k+1} t^2}{2\Lambda_R \prod_i \Lambda_{r_i}} = \frac{\theta(d) t^2}{2R} \prod_i a_i \leq \frac{\theta(d) t^2}{2R^2} k (2/3)^{k-1} \leq \frac{2\theta(d)}{3} \left(\frac{t}{t-1}\right)^2 < 1.$$

The last inequality holds for $t \geq 8$; for $t = 4$, use $\theta(d) = 64/85$. Their sign-disjointness refinement is unavailable in characteristic two, since $-y = y$. Their complete 3-primitive classification [25, Theorem 1.6] does imply positivity in the present $s = 2$ case, by taking base $E = \mathbb{F}_{16}$ and degree two. It does not classify the growing index $t-1$.

The primitive-point framework of [26] is also relevant. Its Fermat-hypersurface count, after primitive lifting, includes two prescribed-order elements with prescribed sum. It does not impose the additional conjugacy $b = a^{t^2}$ that defines M_c. For example, the adjacent-pair counts at $t = 8, 32$

are $24,240$, respectively, while $M_1 = 0$ in both fields. Its Dwork-regular estimate requires degree prime to the characteristic; the natural norm polynomial here has degree four in characteristic two. Its exact hyperplane formula assumes an odd Fermat-prime field. These results do not directly give the four-point bound or the exceptional relative-trace groups.

The rational-image framework of [27] contains these counts as well, including its prescribed inverse trace. Over base E, put $q = t^2$, $m = 2$ and

$$F_c(X) = \frac{X}{X + c}, \qquad r_1 = t - 1, \quad r_2 = t^2 - 1, \quad f_1 = f_2 = 1.$$

Both numerator and denominator are irreducible, coprime linear polynomials, and the denominator is not divisible by X. On the trace-c slice, $F_c(a) = a^{1-t^2}$. If a has order d, then $F_c(a)$ has order B, and both are normal over E. Let $\mathcal{C}_c(b)$ be their count $C_{F_c,c,b}(d, B, 1, (X-1)^2)$ with these parameters. The two auxiliary elements are uniquely determined because $f_1 = f_2 = 1$. Since $\mathrm{Tr}_{L/E}(a^{-1}) = c\, a^{-B}$, we have exactly

$$M_c = \sum_{\substack{w \in U \\ \mathrm{ord}(w) = A}} \mathcal{C}_c(cw).$$

Here $X^2 - 1 = (X - 1)^2$, so their factor coprime to $X - 1$ is 1 and the remaining additive squarefree-divisor factor is 2. Their Corollary 3.2, equation (3.9), would require

$$t^{-2} > 10(t-1)^2(t+1)W(d)W(B),$$

which is impossible. Their Theorem 3.3 retains the same left side for every sieve core, while its right side is at least $5(t-1)(t^2-1)$: its sieve factor $\Delta \geq 1$. More generally, all their displayed general and sieve criteria have left side $q^{m/2-k_1-k_2-2} \leq 1$ when $m \leq 4$, and right side at least $n_1 + 3n_2 + 1 \geq 2$. Thus using base K and $m = 4$ cannot repair the bound; it also prescribes only the coarser trace $c + c^t$. Their Lemma 3.5 and Theorem 4.3 require $m > 2(k_1 + k_2 + 2)$, and their numerical section treats $m \geq 7$. These comparisons retain the prior counting framework without attributing the present effective estimate or exceptional groups to its stated results.

The prescribed-norm framework of [21] also contains these counts. Put $m_{b,c} = \#\{a \in L : \mathrm{ord}(a) = d, N_{L/E}(a) = b, \mathrm{Tr}_{L/E}(a) = c\}$. Then $M_c = \sum_{\mathrm{ord}(b)=A} m_{b,c}$, and each $m_{b,c}$ is a specialization of their Section 4 count: for a fixed primitive lift $\beta \in E$ with $\beta^{t-1} = b$, the power map has a unique preimage with norm β, since $\lambda \mapsto \lambda^2$ permutes the $(t-1)$st roots of unity. Their effective conditions do not yield positivity here. For the K-valued slices, take base K, $n = 4$, $r = t - 1$, $k = 1$, $g = X - 1$, norm 1 and trace 0; over E, for any $c \in E^*$, take $n = 2$, $r = t - 1$, $k = 0$, $g = 1$, norm b of order A and trace $c \neq 0$. The repeated-factor convention in [21, Theorem 4.1 and Corollary 4.1] gives, respectively,

$$t^{-1} > (t-1)W(d), \qquad t^{-2} > (t-1)W(B),$$

both impossible for $t \geq 4$. The first count also aggregates the nonzero K-valued relative traces. Their Proposition 4.1 fixes r, k and assumes $n \geq 4k + 5$; neither specialization lies in that range. Their Section 4.1 gives large-field sufficient bounds, rather than the exceptional slices classified in Theorem 5.4.

D.3 Primitive-normal aggregates and projective geometry

Hachenberger's earlier quartic theorem is a geometric predecessor [11, Theorem 1 and §3]. For even t it gives

$$PN_4(t) \geq \varphi(t^4 - 1) - \varphi(t-1)\varphi(t^2 + 1).$$

Its projective spread counts primitive points on primitive lines, excluding their trace-zero hyperplane intersections. In our notation the exact bridge to that aggregate is

$$PN_4(t) = \varphi(t-1) \sum_{c \in E \setminus K} M_c.$$

Indeed $L^* = K^* \times C_d$, and every generator of C_d has exactly $\varphi(t-1)$ primitive ambient-field lifts. Scaling by K^* preserves normality. In characteristic two and degree four, normality over K is equivalent to $\mathrm{Tr}_{L/K}(a) \neq 0$, or $c + c^t \neq 0$. Consequently its lower bound is equivalent to

$$\sum_{c \in K^*} M_c \leq \varphi(B).$$

It bounds an aggregate rather than every prescribed trace. For example its Table 1 gives $PN_4(8) = 1512$ and $PN_4(32) = 465000$, while $M_1 = 0$ at both parameters. These aggregate counts are consistent with the two empty trace-one slices and do not classify the exceptional groups. The later norm-one and ovoid model [12] provides further geometric context; Supplementary Section S7 compares its trace sections with the additional norm-to-trace ratio fixed here.

The comparisons in this appendix concern stated results and the explicit specializations above. They credit the established counting frameworks, geometric methods and cyclic-group tools; they do not rule out every possible further refinement of those methods or assert absolute global priority.

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